Weak (Proxy) Factors Robust Hansen-Jagannathan Distance For Linear Asset Pricing Models

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Abstract

The Hansen-Jagannathan (HJ) distance statistic is widely used for model selection and specification testing. We show its misbehavior in the presence of weak (proxy) factors. When (proxy) factors exhibit small correlations with asset returns, model selection based on the HJ distance can favor weak (proxy) factors, and the conventional specification test procedure based on the HJ distance has poor finite sample performance and can be size distorted even in large samples. We provide novel procedures for specification testing which are robust against the presence of weak (proxy) factors, and a robust risk premia estimator. Our simulation exercises support our theory, and empirical applications support a four-factor model for Fama French portfolios.

Keywords: asset pricing; identification robust statistics

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1 Introduction

Linear factor models have gained tremendous popularity in the empirical asset pricing literature, e.g.Fama and French (1993) Lettau and Ludvigson (2001), Kan and Zhou (2004), Kan and Robotti (2008). Although the factor structure lying in the asset returns is well-documented (Kleibergen and Zhan (2015)), the true factors are unobserved. Harvey et al. (2016) list hundreds of papers and factors that attempt to explain the cross-section of expected returns. Since so many (proxy) factors are introduced, the asset pricing models are at best approximations. To evaluate these factors and diagnose the model specifications, the HJ distance, proposed in Hansen and Jagannathan (1997), Jagannathan and Wang (1996), is widely used in empirical asset pricing literature.

Since linear asset pricing models are at best approximations and so many factors are proposed, it is more appealing to determine whether or not a model is rejected by the data, namely how good a model can approximate, than to simply identify important common factors (or significant risk premia). This is where specification tests play a role. However, specification tests can be misleading. In a empirical relevant setting where proxy factors are weakly correlated with the unobserved common factors, we show that the HJ specification test can be size distorted even in large samples. It is well documented that lack of model identification can lead to spuriously significant risk premia (Kleibergen (2009),Bryzgalova (2016), Anatolyev and Mikusheva (2018)), and we show that it also affects specification tests with respect to whole models. The boundary, determined via the HJ specification test, between correct model specifications and misspecifications begins to blur in these so-called weak identification cases. Another potential problem would be omitted strong factors. The resulting strong cross-sectional dependence in the error term can exaggerate all sorts of distortions when some included (proxy) factors are weak (Kleibergen and Zhan (2015)).

One of reasons for these failures is that sampling errors are no longer negligible asymptotically in the presence of weak (proxy) factors. Therefore, conventional asymptotic justification may fail in empirically relevant settings, since weak (proxy) factors are commonly observed

in many recent studies (Kleibergen (2009),Anatolyev and Mikusheva (2018)). Gospodinov et al. (2017) discuss the potential power loss of the $\mathcal{J}$ specification test when useless factors are present. We mainly focus on the weak (proxy) factors that are minorly correlated with asset returns instead of the completely useless factors (Gospodinov et al. (2017)), and the size performance of the HJ specification test. We provide two alternative estimators of the HJ distance, and propose two novel specification test procedures which remedy the size distortion.

We also show that similar to the second pass R^2 studied in Kleibergen and Zhan (2015), the HJ statistic can be a misleading gauge of the model fit and is not a satisfactory model selection tool.

Furthermore, we contribute to the literature on the estimation of risk premia in the presence of weak factors and omitted factors. For linear asset pricing models, the conventional approach for estimating risk premia is known as the Fama-Macbeth (FM) two-pass estimation procedure (Fama and MacBeth (1973)), where risk premia estimates result from regressing average asset returns over first-pass estimated risk exposures (factor loadings βs). The two-pass procedure is easy to implement, but can result in unreliable estimates and inference when some included factors are not strongly correlated with asset returns such that their risk exposures do not dominate corresponding sampling errors (Kleibergen (2009), Anatolyev and Mikusheva (2018)). This resembles the failure of the 2SLS estimator in instrumental variable regression when instruments are weak. Besides, Anatolyev and Mikusheva (2018) show that the missing factor structure exacerbates the weak (proxy) factor problem.

Recent paper have developed different estimation techniques to incorporate some of these aforementioned issues. Bryzgalova (2016) provides an estimation approach using shrinkage-based dimension-reduction technique which excludes weak/useless (proxy) factors. Anatolyev and Mikusheva (2018) propose an estimation procedure based on sample-splitting instrumental variables regression with proxies for the missing factor structure. Giglio and Xiu (2017) propose a three-pass estimation procedure and bypass the omitted factors bias

by projecting risk premia of observed factors on those of strong factors extracted via principal components analysis. Another direction is to apply identification robust test statistics to correct for the overly optimistic statistical inference of the risk premia (e.g. Kleibergen (2009), Kleibergen and Zhan (2019), Kleibergen et al. (2018)).

We propose a new risk premia estimator that is robust to the aforementioned issues. Our procedure is easy to implement with existing econometric tools, and provides consistent results for the included factors in an asymptotic framework where the number of assets is comparable to that of the observation periods.

The paper is organized as follows: Section 2 reviews the basic model setting and and shows that conventional procedures are not robust to weak (proxy) factors. Section 3 and 4 introduce our alternative estimators for the HJ distance, and propose our model specification test procedures. Section 5 discusses simulation exercises and empirical applications.

We use the following notation in this paper: P_X stands for $X(X'X)^{-1}X'$ for a full column rank matrix X, M_X for $I - P_X$, $X^{\frac{1}{2}}$ for the upper triangular matrix from the Cholesky decomposition of the positive definite matrix X such that $X = (X^{\frac{1}{2}})'X^{\frac{1}{2}}$. Besides, in the following discussions, certain notation would be more precise with N, T in the sub- or superscripts. For example to model the weak (proxy) factors, it might be reasonable to use notations such as $\beta_{T,N}, d_{g,T,N}, \gamma_{N,T}, \theta_{g,N,T}$ since the parameter values may change according to the sample dimensions in order to model the local to zero behavior. To avoid a cumbersome notation, we ignore these scripts when there can be no misunderstanding.

2 Models and Problems

In this section, we introduce linear asset pricing models and the conventional model selection and specification test procedure based on the HJ distance. We use the term HJ statistic to denote the conventional squared HJ distance estimator reviewed in this section, and to distinguish it from the other two estimators, our so-called HJN and HJS statistics, proposed in sections 3 and 4. We start by introducing our baseline model setting. We next derive asymptotic properties of the HJ statistic in the presence of weak (proxy) factors to clarify the problems we focus on.

2.1 Baseline model setting

We work with the linear asset pricing model because of its popularity among empirical studies. This model imposes that all asset returns share common risk factors described by a small set of proposed factors. We regard the proposed factors in empirical studies as proxies for latent ones in the form as suggested in (Kleibergen and Zhan (2015)). Assumption 2.1 summarizes the baseline model settings:

Assumption 2.1. *For the $N \times 1$ vector of asset gross returns r_t, we assume that*

$$r_t = c + \beta f_t + u_t, \tag{1}$$

with f_t a $K \times 1$ vector of (possibly) unobserved zero-mean factors and u_t is an $N \times 1$ vector of idiosyncratic components, and with a $K \times 1$ vector of proxy factors g_t

$$f_t = d_g \left(g_t - \mu_g\right) + v_t, \tag{2}$$

where g_t is uncorrelated with u_t, v_t, d_g is of full rank and g_t, v_t, u_t are stationary with finite

fourth moments. Furthermore,

$$c = \iota_N \lambda_0 + \beta \lambda_f, \tag{3}$$

where $\lambda_0 \neq 0$ *is the zero-beta return,* λ_f *is a* $K \times 1$ *vector of risk premia, and the parameter space of* λ_0, λ_f *is compact.*

Assumption 2.1 describes the beta representation of linear asset pricing models, and the DGP is similar to the one employed in Kleibergen and Zhan (2015). The moment conditions (or in other words the structural assumptions imposed on the constant term), $c = \iota_N \lambda_0 + \beta \lambda_f$, are commonly used in linear asset pricing model (e.g. Cochrane (2009)). If f_t is observed, then we would have perfect proxies with $d_g = I_K$, $v_t = 0$. Therefore, this model setting also embeds the model specification used in e.g. Kleibergen (2009), Anatolyev and Mikusheva (2018) where factors f_t are assumed to be observed. Using the observed factors g_t, model (1) can be rewritten as

$$r_t = c + \beta_g \bar{g}_t + \left(u_{g,t} + \beta_g \left(\bar{g} - \mu_g\right)\right), \tag{4}$$

with $\beta_g = \beta d_g, \bar{g}_t = g_t - \bar{g}, \bar{g} = \sum_{t=1}^{T} g_t/T, u_{g,t} = \beta v_t + u_t$. The estimation of the risk premia λ_g is usually accomplished by the FM two-stage estimator (Fama and MacBeth (1973), Shanken (1992)). In the first stage, the risk exposures β_g are estimated by regressing asset returns r_t on a constant and factors $\bar{g}_t$, and in the second stage the FM estimator $\widehat{\lambda}_g$ results from regressing average asset returns $\bar{r} = \sum_{t=1}^{T} r_t/T$ on an $N \times 1$ unity vector ι_N and the risk exposure estimates $\widehat{\beta}_g$.

Another well-known representation of asset pricing models is the stochastic discount factor (SDF) representation, based on which the HJ distance is defined. Cochrane (2009) shows that for linear asset pricing models, there is a corresponding SDF m_t that is linearly

spanned by the latent factors

$$m_t(\theta) = F_t'\theta, \tag{5}$$

with $F_t = (1, f_t')'$, and re-scaled risk premia $\theta = (1/\lambda_0, -V_{ff}^{-1}\lambda_f/\lambda_0)$. The moment conditions (3) are then equivalent to the following ones

$$\mathbb{E}\left(m_t(\theta_0)r_t\right) = \iota_N, \tag{6}$$

with ι_N an $N \times 1$ unity vector with all entries equal to one. The population pricing errors which are the deviations from the moment conditions (6) is denoted by

$$e(\theta) = \iota_N - \mathbb{E}\left(m_t(\theta)r_t\right) = \iota_N - q\theta, \tag{7}$$

with $q = \mathbb{E}\left(r_t F_t'\right)$ a full column rank $N \times K$ matrix.

To evaluate the goodness-of-fit of a asset pricing model with given proxy factors, Hansen and Jagannathan (1997) propose the minimum distance δ between the SDF of an asset pricing model and a set of correct SDFs as a measure of model misspecification. It also serves as a measure of goodness-of-fit. A smaller value of the HJ distance indicates a better model fit, and this is used for model selection. The population squared HJ distance has an explicit expression

$$\delta^2 = \min_{\theta} e(\theta)Q_r^{-1}e(\theta) = \iota_N'\left(Q_r^{-1} - Q_r^{-1}q\left(q'Q_r^{-1}q\right)^{-1}q'Q_r^{-1}\right)\iota_N, \tag{8}$$

with $Q_r = \mathbb{E}\left(r_t r_t'\right)$ a full column rank $N \times N$ matrix. After some simple algebra we can rewrite the squared HJ distance as $\iota_N'(Q_r^{-1} - Q_r^{-1}B(B'Q_r^{-1}B)^{-1}B'Q_r^{-1})\iota_N$ with $B = (c, \beta)$, and it is zero if and only if moment conditions (6) hold. Given the observed proxy factors

g_t, the sample counterpart of the squared HJ distance, the HJ statistic, is

$$\widehat{\delta}_g^2 = \min_{\theta_g} \delta_{g,T}^2(\theta_G) = \iota_N' \left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} q_{G,T} \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \right) \iota_N, \tag{9}$$

with $\delta_{g,T}^2(\theta_G) = e_{g,T}(\theta_G)' \widehat{Q}_r^{-1} e_{g,T}(\theta_G), e_{g,T}(\theta_G) = \sum_{t=1}^T e_{g,t}(\theta_G)/T = \iota_N - q_{G,T}\theta_G, e_{g,t}(\theta_G) = \iota_N - r_t G_t' \theta_G, q_{G,T} = \sum_{t=1}^T r_t G_t'/T, \widehat{Q}_r = \sum_{t=1}^T r_t r_t'/T$, $G_t = (1, \bar{g}_t')'$. The estimator resulting from this quadratic optimization problem is

$$\widehat{\theta}_G = \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \iota_N \tag{10}$$

Jagannathan and Wang (1996) propose the HJ specification test using the HJ statistic. They test the model specification by testing the moment conditions (6). Under the null hypothesis that the moment conditions (6) hold, the critical values for this non-pivotal test statistic are drawn from the weighted sum of independent $\chi^2(1)$ random variables

$$\sum_{i=1}^{N-K-1} p_i x_i, \tag{11}$$

with x_i being independent $\chi^2(1)$ distributed random variables, and p_i being the positive eigenvalues of the matrix

$$\widehat{S}^{\frac{1}{2}} \left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} q_{G,T} \left(q_{G,T}' \widehat{Q}_r^{-1} q_{G,T} \right)^{-1} q_{G,T}' \widehat{Q}_r^{-1} \right) S_T^{\frac{1}{2}'},$$

with $\widehat{S} = S_T(\widehat{\theta}_G), S_T(\theta_G) = \frac{1}{T} \sum_{t=1}^T e_{g,t}(\theta_G) e_{g,t}(\theta_G)', e_{g,t}(\theta_G) = \iota_N - r_t G_t' \theta_G$.

2.2 Problems and asymptotic properties

Our interest lies in the performance of the HJ statistic in the presence of weak identification issue, in particular, when observed proxies g_t are only weakly correlated with asset returns. Ahn and Gadarowski (2004) document the poor finite sample performances of the HJ spec-

ification test, and they argue that the size distortion is due to the critical value of the test which requires the estimation of the covariance matrix of $e_t(\theta)$ that performs badly with a limited number of observation periods. This is also consistent with the findings in Kleibergen and Zhan (2019), Kleibergen et al. (2018). In later parts, we show that not only in finite samples but also in large samples the HJ specification test can be severely size distorted in the presence of weak identification issues. We focus on *two issues* that can cause the potential deficiency of the HJ statistic.

1. Weak (proxy) factors. The HJ statistic depends on the estimator $\widehat{\theta}_G$. This is a GMM estimator based on the moment conditions (6) with weighting matrix $\widehat{Q}_r$. Similar to the FM risk premia estimator, this estimator can be constructed in two stages. In the first stage q_g is estimated via the estimator $q_{G,T}$, and in the second stage $\widehat{\theta}_G$ results from regressing $\widehat{Q}_r^{-\frac{1}{2}}\iota_N$ on the first stage estimates $\widehat{Q}_r^{-\frac{1}{2}}q_{G,T}$. This close link with the FM estimator raises concerns with respect to the quality of the θ_G estimator.

The FM estimator are highly unreliable under weak identification (e.g. Kleibergen (2009), Kleibergen and Zhan (2015), Kleibergen and Zhan (2019), Kleibergen et al. (2018), Anatolyev and Mikusheva (2018)). For linear asset pricing models, the identification strength is reflected by the rank of $B_g = (c, \beta_g)$ (e.g. Kleibergen and Zhan (2019)). The weak identification issues result from the empirical observation that the matrix B_g might be of reduced rank or near reduced rank. For example, this can happen when some (proxy) factors used in the estimation are weakly correlated with the asset returns. Suppose the β_g matrix is small, for example of order $O(1/\sqrt{T})$, and then the sampling errors in the first stage estimator $\widehat{\beta}_g$ are of the same order, $O_p(1/\sqrt{T})$, as the signal. The sampling errors in the first stage estimator $\widehat{\beta}_g$ are then no longer negligible, and this leads to the asymptotic invalidity of the FM estimator.

Following the same reasoning, the asymptotic justification for the estimator $\widehat{\theta}_G$ fails when q_g is small and comparable to its sampling error (see Theorem A.2). Since $q_g = \beta_g V_g$ with $V_g = \mathbb{E}g_t g_t'$, we model weak (proxy) factors using drifting to zero risk exposures (see

Assumption 2.3) to mimic the behavior of a small q_g, which is in line with the literature on weak factors (e.g. Kleibergen (2009)).

2. The missing factor structure. Omitted factors have drawn attention in recent studies, e.g. Giglio and Xiu (2017). Equation (4) suggests that when we work with observed factors g_t in a latent factor setting, the omitted factors v_t contribute to the error term $u_{g,t}$. We allow for a missing factor structure in u_t such that $u_t = \gamma z_t + e_t$. Namely, we allow that there are unobserved and unpriced factors z_t that explain most of the cross-sectional dependence in u_t (Assumption 2.1, 2.2). Anatolyev and Mikusheva (2018) show that in the beta representation of the linear asset pricing model, the presence of an unobserved (missing) factor structure in the error terms creates the classical omitted-variables problem in the second stage regression of the FM two-stage risk premia estimator and thus leads to a bias in the estimates when the number of assets is large. When working with the SDF representation, we encounter a similar issue that the missing factor structure could exacerbate the problem caused by the weak (proxy) factors and enlarges the bias in the estimator $\widehat{\theta}_g$ (see Theorem A.2).

Therefore, when above *two issues* are present, the HJ statistic uses an estimate that is potentially far away from the true value, and thus selection and inference based on the HJ statistic can be misleading. Before we continue to verify this, we make two assumptions.

Assumption 2.2. *Suppose u_t can be decomposed into two parts: a missing factor structure with a $K_z \times 1$ ($K_z \geq 0$) vector of unobserved strong factors z_t and weakly cross-sectional correlated noise e_t;*

$$u_t = \gamma z_t + e_t, \tag{12}$$

where (i) e_t are independent from $e_s, s \neq t$ and $g_{t'}, \forall t'$ with mean zero and bounded fourth moments $\sup_i \mathbb{E} e_{it}^4 < L < \infty$; (ii) denote $\Omega_e = \mathbb{E} e_t e_t'$, then $\lim_{N,T} tr(\Omega_e)/N = a > 0$ and $0 < l < \liminf_{N,T} \lambda_{\min}(\Omega_e) < \limsup_{N,T} \lambda_{\max}(\Omega_e) < L < \infty$ with $\lambda_{\min}(X)$ the smallest eigenvalues of matrix X and $\lambda_{\max}(X)$ the largest eigenvalues of matrix X; (iii)

$\mathbb{E}\left|\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left(e_{it}^2-\mathbb{E}e_{it}^2\right)\right|^4 < L < \infty.$

Assumption 2.3. *Denote Q_r by the second moments of r_t, $\eta = \left(\eta_{B_g}, \gamma\right)$ with $\eta_{B_g} = \left(c, \eta_{\beta_g}\right), \eta_{\beta_g} = \left(\beta_{g,1}, \sqrt{T}\beta_{g,2}\right), \beta_{g,i} = \beta d_{g,i}, i = 1,2$ being of dimension $N \times K_{g,i}, i = 1,2$ $(K_{g,1} + K_{g,2} = K, K_{g,2} \geq 0)$ and full column rank matrices. (i) for fixed N, we assume $\eta'\eta$ is a $(1+K+K_z)\times(1+K+K_z)$ positive definite matrix; (ii) as N, T approach to infinity, $N^{-1}\eta'\eta$ converges to a $(1+K+K_z)\times(1+K+K_z)$ positive definite matrix.*

Our framework involves the observed factors g_t, and the omitted ones v_t, z_t. They have factor loadings β_g, β, γ respectively. Assumption 2.3 specifies the strengths of these factors. The loadings, $\beta_{g,2}$, with respect to $K_{g,2}$ proxy factors $g_{2,t}$ are modeled as drifting to zero sequences, so we call $g_{2,t}$ weak proxy factors. We do not restrict the strength of the priced latent factors f_t, and allow for weak priced latent factors. This assumption resembles the factor loading assumption in Anatolyev and Mikusheva (2018), but our risk exposure matrix (β_g, β, γ) is of reduced rank.

Assumption 2.2, which is similar to assumptions in Onatski (2012) and Anatolyev and Mikusheva (2018), does not fully rule out the cross-sectional dependence in the idiosyncratic error term e_t. This assumption allows the explanatory power of the cross-sectional variation in e_t to be comparable to the weak proxy factors when N,T increase proportionally, and the weak identification issue rises up when the explanatory power of the proxy factors are roughly of the same order as the e_t. When the cross sectional size N is fixed, the assumptions 2.2.(ii)-(iii) imposed on the noise term e_t hold naturally as long as Assumption 2.2.(i) holds and we can not really distinguish weak and strong factors. In later parts when N is fixed, we do not make use of the assumptions imposed on e_t but only the independent assumption (Assumption 2.2.(i)). We assume the independence across periods which is consistent with the efficient market hypothesis, and given empirical studies mostly use monthly or even less frequent data this is not a very wild assumption.

Lemma 2.1. *Suppose Assumption 2.1, A.1 hold, let T increase to infinity then*

$$\widehat{B}_g =_d (c, \beta_g) + \psi_{B_g}/\sqrt{T}$$

where $\widehat{B}_g = q_{G,T}\widehat{Q}_G^{-1}, \widehat{Q}_G = \sum_{t=1}^{T} G_t G_t'/T, \psi_{B_g} = \left(\psi_c, \psi_{\beta_{g,1}}, \psi_{\beta_{g,2}}\right)$ *and* $vec(\psi_{B_g})$ *being zero-mean normal random vectors.*

Proof: see Appendix A.1.

Theorem 2.2. *Suppose Assumption A.1 and 2.1-2.3 hold, let T increase to infinity with fixed N then the behavior of the HJ statistic is characterized by:*

$$T\widehat{\delta}_g^2 \to_d \widetilde{\psi}_{B_g}' M_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g} + (0 \vdots \psi_{\beta_{g,2}})\right)} \widetilde{\psi}_{B_g}$$

with $\widetilde{\psi}_{B_g} = \psi_{B_g} Q_G \theta_G$.

Proof: see Appendix A.1.

Theorem 2.2 is derived assuming that the linear model is correctly specified, and it simply suggests that with strong factors, $K_{g,2} = 0$, the weighted sum of χ^2's provides a reasonable approximation (Corollary 2.2.1).

Corollary 2.2.1. *Suppose the assumptions in Theorem 2.2 hold and $K_{g,2} = 0$, then*

$$T\widehat{\delta}_g^2 \sim_d \sum_{i=1}^{N-K-1} p_i x_i$$

with x_i being independently $\chi^2(1)$ distributed random variables and p_i being the positive eigenvalues of the matrix $S_T^{\frac{1}{2}}\left(\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1} q_{G,T}\left(q_{G,T}'\widehat{Q}_r^{-1} q_{G,T}\right)^{-1} q_{G,T}'\widehat{Q}_r^{-1}\right) S_T^{\frac{1}{2}\prime}$.

Proof: this is a direct result of Theorem 2.2.

However, the conventional specification test procedure can be unreliable and suffer from severe size distortion even in large samples due to the irregular distribution of the HJ statistic (Corollary 2.2.2).

Corollary 2.2.2. *Suppose Assumption A.1 , A.2 and 2.1 - 2.3 hold with* $0 < K_{g,2} \leq K$,

$$\limsup_{N} \limsup_{T} \quad \mathbb{P}\left(T\widehat{\delta}_g \geq \widehat{c}_{1-\alpha}\right) = 1$$

where $\widehat{c}_{1-\alpha}$ *is the conventional critical value derived from the distribution (11).*

Proof: see Appendix A.2.

Corollary 2.2.2 shows that under certain conditions the conventional specification test reject the model specification with probability converging to one even when the moment conditions hold. Thus the conventional specification testing procedure based on the HJ statistic may mistake the "weak identification" resulting from the weak (proxy) factors for the model misspecification, and leads to over-rejection when models are correctly specified.

2.3 Simulation exercises

We conduct simulation exercises to show that the HJ distance statistic as a model selection criterion might favor the presence of useless factors, and the HJ specification test suffers from severe size distortions.

In the first simulation exercise (Figures 1, 2), we calibrate the data generating process to match the data set of asset monthly gross returns on 25 size and book to market sorted portfolios from 1963 to 1998 and the three Fama French (FF) factors used by Lettau and Ludvigson (2001). The data is simulated in the following way: we simulate three proxy factors $g_t \sim \ i.i.d\ N(0, V_F)$, three omitted factors $v_t\ i.i.d\ \sim N(0.99V_F)$ and three strong factors are then generated by $f_t = 0.1 * g_t + v_t$,. V_F is calibrated to the sample covariance of the FF factors. We also generate three completely useless factors $w_t \sim i.i.d.N(0, V_F)$. We then generate returns via $r_t = c + \beta f_t + u_t$, $u_t \sim \ i.i.d\ N(0, V_u)$, where c is the sample mean of the returns from the data, β is the sample slope parameter between the assets returns and FF factors from the data. and V_u is the sample covariance of the residuals resulting from regressing asset returns on a constant and FF factors from the data.

Figure 1 compares the density functions of the simulated HJ statistics evaluated with various combinations of the factors g_t, f_t, w_t. For example, the black solid curve is drawn using three strong factors f_t, the black dashed curve is drawn with two strong factors f_{1t}, f_{2t}. Ideally, the black solid curve should on the most left, then the model with three strong factors is most likely to be selected via the HJ statistic. However, comparing the red solid and black solid curves shows that adding additional useless factors leads to a shift of the distribution to the left, so it reduces the HJ statistic and leads to a "preferred model". This is similar to the case where additional useless regressors can increase the cross-sectional R^2. The blue sold curve illustrates the density function of the HJ statistic of the model with three weak proxy factors. By construction, the moment conditions (6) are satisfied by the three weak proxy factors, and this model is correctly specified. If we compare the blue solid curve with the blacked dashed one which is constructed with only two strong factors, then the misspecified model with two strong factors is more likely to be selected. Similar to the findings in Kleibergen and Zhan (2015) where they show that the second stage R^2 in linear asset pricing models is unreliable when weak/useless factors are included, the HJ statistic is not a satisfying model selection tool.

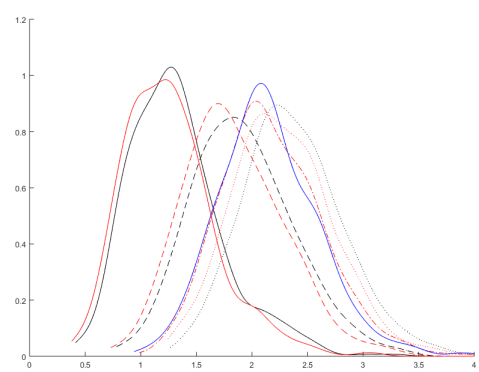

Figure 1: Density functions of the HJ statistic: (1) black solid: three strong factors; (2) black dashed: two strong factors; (3) black dotted: one strong factors; (4) blue solid: three weak proxy factors; (5) red solid: three strong factors and one useless factor; (6) red dashed: two strong factors and one useless factor; (7) red dotted: one strong factors and one useless factor; (8) red dash-dotted:one strong factors and two useless factors

Figure 2 compares two different approaches for approximating the distribution of the HJ statistic: one uses the weighted sum of χ^2's, and another one uses the infeasible distribution from Theorem 2.2. The left-hand side panels of Figure 2 use three strong factors, while the right-hand side panels use three week ones.

The upper panels of Figure 2 show that both approximations are bad when T is small, and shift to the left compared with the density function of the HJ statistic. This is consistent with the observation in Ahn and Gadarowski (2004) that the HJ specification test over-rejects correct model specifications in small samples. With a limited number of observation periods, not only sampling errors in the q_G estimators need to be taken into account but also those of other estimators such as the covariance estimator S_T (e.g. Kleibergen and Zhan (2019), Kleibergen et al. (2018)). The infeasible distribution improves slightly by taking into account the sampling errors in the q_G estimators.

When T is large, the randomness in the covariance estimators becomes small, but the sampling errors in $q_{G,T}$ still matter when proxy factors are weak. As shown in lower panels of Figure 2, with a larger sample size the conventional approximation works fine when factors are strong but not when weak proxy factors are present. With weak proxy factors, the distribution of the HJ statistic is not properly approximated by the weighted sum of χ^2's even in large samples, and the HJ specification test is still likely to over-reject models when moment conditions do hold. This is consistent with Corollary 2.2.2.

Our second simulation exercise considers a simple single factor model in order to further illustrate the size distortion of the HJ specification test. We calibrated parameters to the data set from Kroencke (2017). We simulate the proxy factor $g_t \sim \ i.i.d.\ N(0, V_f/4)$, the omitted factor $v_t \sim \ i.i.d.\ N(0, Vff - d_g * (V_f/4) * d_g')$, the latent factor $f_t = d_g g_t + v_t$ and thus the variance of the f_t remain unchanged to different values of the d_g. We calibrate V_f to match the sample variance of the consumption growth factor from the data. The factor proposed in Kroencke (2017) has been shown to be weak (e.g. Kleibergen and Zhan (2019)). Therefore, we choose β to be of $10\widehat{\beta}$ with $\widehat{\beta}$ the sample regression parameter from

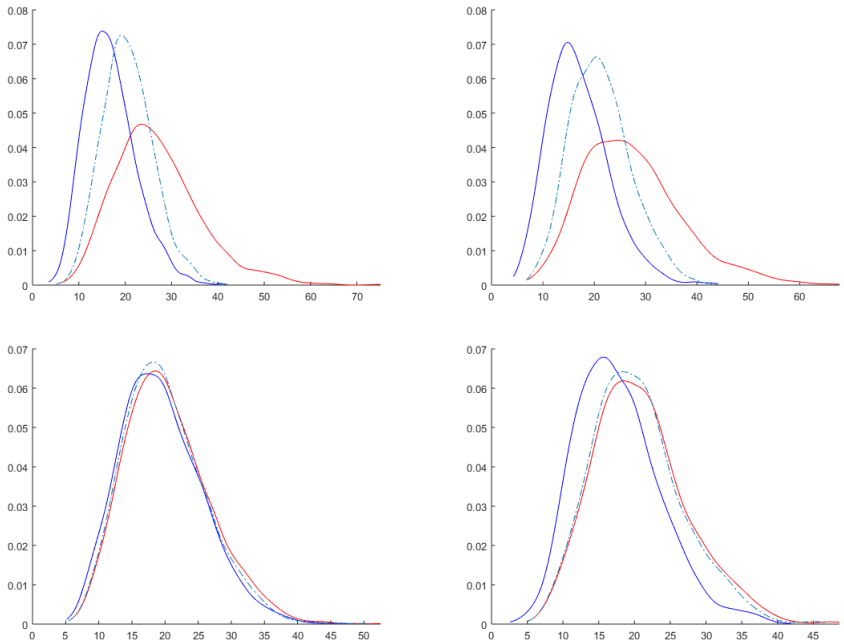

Figure 2: Density functions of the HJ statistic (red solid curve), weighted sum of χ^2's approximation (blue solid curves), asymptotic distribution based on Theory 2.2 (dot-dashed curve). Top left-hand side panel: $T = 100$, three strong factors f_t; top right-hand side panel: $T = 100$, three weak factors g_t; bottom left-hand side panel: $T = 1000$, three strong factors f_t; bottom right-hand side panel: $T = 1000$, three weak factors g_t.

Kroencke (2017), and thus r_t is generated with one single strong factor f_t. We arbitrarily choose $d_g = 1.9; 0.9$ to mimic a strong and weak proxy factor. Table 1 shows that the HJ specification tests have poor finite sample performances, size distortions increase with the number of assets and with relative weak proxy factors the distortion is more severe. These observations are consistent with Corollary 2.2.2.

β	$N = 5$	$N = 10$	$N = 15$	$N = 31$
T=100, $d_g = 1.9$	0.5032	0.8824	0.9711	0.9992
T=10000, $d_g = 1.9$	0.1210	0.1486	0.2298	0.2238
T=100, $d_g = 0.9$	0.7132	0.9330	0.9814	1
T=10000, $d_g = 0.9$	0.5174	0.8906	0.8834	0.9978

Table 1: Rejection frequency table under the null (at 0.05 significance level) via the HJ specification test with critical values drawn from weighted sum of χ^2's

3 Specification testing with limited N: HJS statistic

We propose the following HJS statistic:

$$\widehat{\delta}_g^* = \inf_{\theta \in CS_{r,\alpha_1}} \delta_{g,T}(\theta), \tag{13}$$

with CS_r the robust confidence set constructed by inverting the AR test statistic:

$$CS_{r,\alpha_1} = \{\theta \in \Theta : AR(\theta) \leq c_{1-\alpha_1}\}; \quad AR(\theta) = Te_T(\theta)' \left(S_T(\theta)\right)^{-1} e_T(\theta). \tag{14}$$

The HJS statistic is constructed in two stages: in the first stage we derive an identification robust confidence set for θ_G; and in the second stage we make use of the identification robust confidence set to construct our test statistic and its critical value. In this way, we combine a less powerful but robust statistic with a non-robust one to incorporate the model identification strength in our testing procedure, and in later discussion we show that this test improves performances in both size and power. This approach can be extended to more general cases.

Before we proceed to show the size and power performances of the HJS specification test (Theory 3.2 and Theory 3.3), we first discuss the properties of the robust confidence set $CS_{r,\alpha}$. We use an Anderson-Rubin (AR) type test statistic (e.g. Gospodinov et al. (2017), Kleibergen (2009)) to construct the robust confidence set of the θ_G. Kleibergen and Zhan (2019) study a similar robust risk premia confident set using the GRS-FAR statistic. They show this kind of set can be unbounded in certain cases. Therefore, for practical reason we restrict the parameter space Θ to be a compact set (Assumption 2.1), of which the robust confidence set is a subset. By construction, when the model is strongly identified we would expect the confidence set to shrink to a singular set as sample size grows, and when the model is weakly identified or even unidentified the diameter of this set can be arbitrarily large. Lemma 3.1 shows that the confidence set cover the true value with requested probability

which is essential for the correct size performance. To complete the construction of the HJS statistic we set $\widehat{\delta}_g^* = \infty$ when CS_{r,α_1} is empty.

Lemma 3.1. *Suppose the assumptions in Theorem 2.2 hold, then*

$$\liminf_T \mathbb{P}\left(\theta_G \in CS_{r,\alpha}\right) \geq 1 - \alpha$$

proof: See the appendix.

Lemma 3.1 implies that the confidence set cover the true value consistently in the presence of weak factors. This result holds under more general cases, for example it holds even when the model is not identified. It is also related to a widely studied weak instrument problem: tests without assuming identifications (Kleibergen (2005)). Lemma 3.1, the correct coverage probability of the confidence set, directly leads to the correct size of this test (Theorem 3.2).

Theorem 3.2. *Suppose the assumptions in Theorem 2.2 hold,*

$$\limsup_T \mathbb{P}\left(T\widehat{\delta}_g^* \geq c_{1-\alpha}^*\right) \leq \alpha$$

where $c_{1-\alpha}^* = \sup_{\theta \in CS_{r,\alpha_1}} c_{1-\alpha_2}^*(\theta)$, $c_{1-\alpha}^*(\theta)$ *is the* $1-\alpha$ *quantile of the weighted sum of* N $\chi^2(1)$ *random variables with weights being the non-zero eigenvalues of* $(\widehat{\Sigma}_e(\theta))^{\frac{1}{2}}\widehat{Q}_r^{-1}(\widehat{\Sigma}_e(\theta))^{\frac{1}{2}'}$ *and* α_1, α_2 *is chosen such that* $(1-\alpha_1)(1-\alpha_2) = 1-\alpha$.
proof: See the appendix.

Theory 3.2 provides a upper bound for the HJS statistic and shows the HJS specification test is size correct in the presence of weak (proxy) factors. The proof of it implies that the size property of the HJS specification test is a direct result of Lemma 3.1, and given the lemma holds on more general conditions we know the HJS specification test can be extended to more general cases as well.

By construction, the HJS specification test is conservative as we take the infimum over the whole confidence set. We use a simple example to show that using the supreme instead

of the infimum can lead to size distortion again.

Example 1. *Suppose Assumptions 2.1, - 2.3 hold. In addition, assume that ψ_{B_g} in Lemma 2.1 satisfies that the $\psi_{B_g} \sim N(0, Q_G \otimes \Sigma)$ with $Q_{\widetilde{G}} = \mathbb{E}(G_t G_t')$, $K = K_{g,2} \geq 2$. Then*

$$\limsup_{T} \mathbb{P}\left(\left(\sup_{\theta \in CS_{r,\alpha_1}} \delta_{g,T}(\theta)\right) \geq c^*_{1-\alpha}\right) = 1$$

proof: See the appendix.

Even though it is a conservative test, the HJS specification test has a better power performance compared with another well-know specification test, the $\mathcal{J}$ specification test. The $\mathcal{J}$ specification test statistic is also constructed based on the AR statistic such that

$$\mathcal{J} = \inf_{\theta} AR(\theta)$$

Gospodinov et al. (2017) show that the $\mathcal{J}$ specification test is size correct in the presence of spurious/useless factors which means q_G is of reduced rank and the model is not identified, but it has a complete power loss in such case. We extend their results to weakly identified models. Theory 3.3 shows that in both cases, in the presence of spurious/useless (proxy) factors and in the presence of weak (proxy) factors, the $\mathcal{J}$ specification test is not consistent while the HJS specification test is.

Theorem 3.3. *Suppose Assumptions A.1, 2.1, 2.2 hold, but instead of the correct proxy factors g_t, wrong proxy factors $\widetilde{g}_t$ are used such that $\widetilde{g}_t$ is a $\widetilde{K} \times 1$ vector, $\left\|\iota_N - q_{\widetilde{G}}\theta_{\widetilde{G}}\right\| > a > 0, \forall \theta_{\widetilde{G}} \in \Theta$ with $q_{\widetilde{G}} = \mathbb{E}(r_t \widetilde{G}_t'), \widetilde{G}_t = (1, \widetilde{g}_t)$ and the model is misspecified. In addition, assume that the ψ_{B_g}, when we replace G_t with $\widetilde{G}_t$, in Lemma 2.1 satisfies that $\psi_{B_g} \sim N(0, Q_{\widetilde{G}} \otimes \Sigma)$ with $Q_{\widetilde{G}} = \mathbb{E}(\widetilde{G}_t \widetilde{G}_t')$. Let $H = \left(\iota_N, q_{\widetilde{G}}\right)$.*

(i) (Gospodinov et al. (2017) Theorem 2, unidentified model under misspecification) Sup-

pose H has a column rank $K+1-k$ for an integer $k \geq 1$, then we have

$$\mathcal{J} \to_d w_{k,i}$$

where $w_{k,i}$ is the smallest eigenvalue of $\mathcal{W}_k \sim \mathcal{W}_k\left(N-K-1+k, I_k\right)$ and $\mathcal{W}_k\left(N-K-1+k, I_k\right)$ denotes the Wishart distribution with $N-K-1+k$ degrees of freedom and a scaling matrix I_k. Furthermore,

$$\limsup_T \mathbb{P}\left(\mathcal{J} \geq c_{\chi^2_{N-K},1-\alpha}\right) \leq \alpha,$$
$$\liminf_T \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) = 1,$$

with $c_{\chi^2_{N-K},1-\alpha}$ the $1-\alpha$ quantile of χ^2_{N-K}.

(ii) (weakly identified model under misspecification) Suppose $(HQ_x)' * (HQ_x)$ with $Q_x = diag(I, \sqrt{T}I_{K_{g,2}})$ converges to a positive definite matrix, then we have

$$\mathcal{J} \to_d w_{k,ii}$$

where $w_{k,ii}$ is the smallest eigenvalue of $\mathcal{W}_k \sim \mathcal{W}_k\left(\mu, N-K-1+k, I_k\right)$ and $\mathcal{W}_k\left(\mu, N-K-1+k, I\right.$ denotes the non-central Wishart distribution with $N-K-1+k$ degrees of freedom and a scaling matrix I_k, a location parameter μ (μ is specified in the proof). Furthermore,

$$\liminf_T \mathbb{P}\left(\mathcal{J} \geq c_{\chi^2_{N-K},1-\alpha}\right) < 1,$$
$$\liminf_T \mathbb{P}\left(T\widehat{\delta}^*_g \geq c^*_{1-\alpha}\right) = 1,$$

with $c_{\chi^2_{N-K},1-\alpha}$ the $1-\alpha$ quantile of χ^2_{N-K}.

proof: See the appendix.

Theorem 3.3 shows the HJS specification test has a better power property than the $\mathcal{J}$

specification test.

3.1 Simulation exercise

Figure 3 illustrates the density function of the HJS statistic under same settings as in Figure 1. We use our previous simulation experiment calibrated to data from Lettau and Ludvigson (2001) to illustrate properties of the HJS statistic. The black dotted curve in Figure 3 suggests the value of the HJS statistic could be small when model is not correctly specified. These smaller values are due to much smaller confidence sets CS_r when model is not correctly specified. Therefore, using the HJS statistic as a model selection tool may not be a good idea even though it provides size correct specification test.

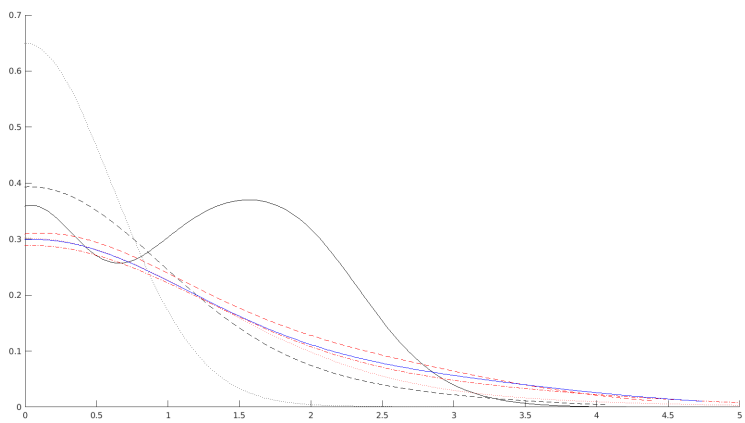

Figure 3: Densities of the HJS statistic: (1) black solid: three strong factors; (2) black dashed: two strong factors; (3) black dotted: one strong factors; (4) blue solid: three weak proxy factors; (5) red solid: three strong factors and one useless factor; (6) red dashed: two strong factors and one useless factor; (7) red dotted: one strong factors and one useless factor; (8) red dash-dotted:one strong factors and two useless factors

4 Specification testing with large N: HJN statistic

In the last section, we construct the HJS statistic using a robust confident set of θ_G instead of identifying its value since θ_G is only weakly identified with limited number of asset returns. The HJS specification testing procedure involves optimization, and in practice it is commonly done through a time consuming grid search procedure. In this section, we provide another novel valid specification test based on the concept of the HJ distance, which does not involve optimization procedure and is easy to implement with existing econometrics tools.

When we work within a dimension-asymptotic framework such that the number of time period observations and that of asset returns grow, weak (proxy) factors do not lead to a weak identification problem (Anatolyev and Mikusheva (2018)). Even though $\widehat{\theta}_G$ is not consistent, another consistent estimator for θ_G can be constructed. This is similar to the case of many weak instruments. This gives another way to construct a test statistic. In the following, we first introduce our θ_G estimator, and then a novel specification test statistic, the HJN statistic, based on it.

Using $\widehat{\theta}_G$ naively in this framework will not lead to consistent estimates (Theory A.2). To resolve the problems of the missing factor structure and the weak (proxy) factors simultaneously, our estimation approach for θ_G makes use of the results from Bai and Ng (2002),Bai (2003) and Giglio and Xiu (2017) to recover the common components in the error terms via principal component analysis, and the instrument variable idea applied for the factor models proposed in Anatolyev and Mikusheva (2018). We propose the following procedure to estimate the θ_G with N base portfolios of gross returns r_t:

Step (1): Estimate $\widehat{c}, \widehat{\beta}_g$ in equation (4) via the simple OLS estimator.

Step (2): Determine the omitted factor structure by the following two steps: (2.1) determine the number of factors, K_{vz}, in $\widehat{u}_{g,t} = r_t - \widehat{c} - \widehat{\beta}_g \bar{g}_t$ by

$$\widehat{K}_{vz} = \arg \min_{0 \leq j \leq K_{vz,\max}} \left(N^{-1} T^{-1} \lambda_j \left(\widehat{u}_g \widehat{u}_g' \right) + j\phi(N,T) \right) - 1$$

where $\widehat{u}_g$ is $T \times N$ matrix stacked with the OLS residuals $\widehat{u}_{g,t}$, $K_{vz,\max}$ is an arbitrary upper bound for K_{vz} and $\phi(N,T)$ is a penalty function with the properties $\phi(N,T) \to 0, \phi(N,T)/(N^{-\frac{1}{2}} + T^{-\frac{1}{2}}) \to \infty$ (In our simulation exercise we simply choose $\phi(N,T) = N^{-\frac{1}{4}} + T^{-\frac{1}{4}}$); (2.2): estimate the $T \times N$ common component matrix $cc = xb'$ stacked with the common components cc_t:

$$\widetilde{cc} = \widehat{x}\widehat{b}, \quad \left(\widehat{x}, \widehat{b}\right) = \arg \min_{b_i, x_t \ s.t. \ \sum_{t=1}^{T} x_t x_t'/T = I_{\widehat{K}_{vz}}} \sum_{i,t} \left(\widehat{u}_{g,it} - b_i' x_t\right)^2$$

The estimator $\widehat{x}$ is equal to the $\sqrt{T}$ times eigenvector associated with the $\widehat{K}_{vz}$ largest eigenvalues of the matrix $\widehat{u}_g \widehat{u}_g'$, and $\widehat{b} = \widehat{x}'\widehat{u}_g/T$ corresponds to the OLS estimator regressing $\widehat{u}_g$ over $\widehat{x}$.

Step (3): Remove the missing factor structure from regressors

$$\widetilde{q}_{G,T}^{(i)} = q_{G,T}^{(i)} - \widetilde{cc}_{G,T}^{(i)'}, i = 1, 2$$

where

$$q_{G,T}^{(1)} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} r_t G_t'; \ q_{G,T}^{(1)} = \frac{1}{T - \lfloor \frac{T}{2} \rfloor} \sum_{t=\lfloor \frac{T}{2} \rfloor + 1}^{T} r_t G_t'$$

$$\widetilde{cc}_{G,T}^{(2)} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} \widetilde{cc}_t G_t'; \ \widetilde{cc}_{G,T}^{(2)} = \frac{1}{T - \lfloor \frac{T}{2} \rfloor} \sum_{t=\lfloor \frac{T}{2} \rfloor + 1}^{T} \widetilde{cc}_t G_t'.$$

Step (4): Our proposed four-pass estimator is then derived via IV regression where we use instrument $\widetilde{q}_{G,T}^{(1)}$ for $\widetilde{q}_{G,T}^{(2)}$ and vise versa:

$$\widetilde{\theta}_G = \sum_{i=1}^{2} \widetilde{\theta}_G^{(i)}/2,$$

with $\widetilde{\theta}_G^{(1)} = \left(\widetilde{q}_{G,T}^{(1)'} P_{\widetilde{q}_{G,T}^{(2)}} \widetilde{q}_{G,T}^{(1)}\right)^{-1} \widetilde{q}_{G,T}^{(1)'} P_{\widetilde{q}_{G,T}^{(2)}} \iota_N$ and $\widetilde{\theta}_G^{(2)} = \left(\widetilde{q}_{G,T}^{(2)'} P_{\widetilde{q}_{G,T}^{(1)}} \widetilde{q}_{G,T}^{(2)}\right)^{-1} \widetilde{q}_{G,T}^{(2)'} P_{\widetilde{q}_{G,T}^{(1)}} \iota_N$.

Compared with the estimator proposed in Anatolyev and Mikusheva (2018), our proposed estimator relaxes the restrictions on the number of omitted factors and the restrictions on the rank of loadings with respect to all the factors present in the model. To illustrate why our proposed procedure is robust against weak (proxy) factors and missing factor structure, we start by comparing it with the conventional θ_G estimator. To do so, we first rewrite equation (6) as

$$\iota_N = q_{G,T}\theta_G - \epsilon_{q_G}\theta_G,$$

with $\epsilon_{q_G} = (q_{G,T} - \mathbb{E}q_{G,T})$ which is correlated with the $q_{G,T}$. The term ϵ_{q_G} is vanishes asymptotically, and so it is dominated by $q_{G,T}$ when all proxy factors are strong. The conventional estimator results from regressing ι_N on $q_{G,T}$ is then valid in large samples, since ϵ_{q_G} becomes negligible. However, if some (proxy) factors are weak, some columns of $q_{G,t}$ are of the same order as ϵ_{q_G}, then there would be a classic endogeneity problem if we simply regress ι_N on $q_{G,T}$.

To solve the endogeneity problem, a valid instrument can be constructed in our framework with a sample splitting technique and this idea is also employed in Anatolyev and Mikusheva (2018). Given the independence of the e_t from non-overlapping sub-samples, $q^{(1)}_{G,T}$ can serve as an instrument for $q^{(2)}_{G,T}$ and vice versa when there is no missing factor structure ($K_{vz} = 0$) and this is the starting point of our proposed procedure. When there is a missing factor structure with factors that might be correlated across time, $q^{(1)}_{G,T}$ is no longer a valid instrument for $q^{(2)}_{G,T}$. Therefore, we use $\widetilde{q}^{(i)}_{G,T}, i = 1, 2$ which results from removing the missing factor structure from $q^{(i)}_{G,T}, i = 1, 2$. By doing so, $\widetilde{q}^{(1)}_{G,T}$ is asymptotically uncorrelated with $\widetilde{\epsilon}^{(2)}_{q_G}$, and is a valid instrument.

As shown in Theorem 4.1, our estimation procedure provides $\min\{\sqrt{T}, \sqrt{N}\}$-consistent results for θ_G.

Theorem 4.1. *Suppose Assumptions 2.1 - 2.3, A.1 - A.11 hold, and $N/T \to c$.*

$$\sqrt{NT}Q_x^{-1}\left(\widetilde{\theta}_G - \theta_G\right) \to O_p(1),$$

with $Q_{B_g,T} = diag(I_{1+K_{g,1}}, \sqrt{T}I_{1+K_{g,1}})$. When $c = 0$, $\widehat{\Sigma}_{\theta_G}^{1/2}\left(\widetilde{\theta}_g - \theta_G\right) \to_d N(0, I)$ with $\widehat{\Sigma}_{\theta_G}$ given in the appendix.

Proof: See the appendix.

Corollary 4.1.1. *Suppose the assumptions in Theorem 4.1 hold, then*

$$\sqrt{NT}Q_x^{-1}\left(\widetilde{\lambda}_g - \lambda_g\right) \to O_p(1),$$

with $\widetilde{\lambda}_g = -V_g diag(0_{K\times 1}, I_K)\widetilde{\theta}_G/\widetilde{\theta}_{G,1}$.

Proof: See the appendix.

Kleibergen and Zhan (2018) study risk premia on mimicking portfolios by projecting non-traded factors on traded base portfolios, and then carry out identification robust tests using a set of testing portfolios. We use a similar idea to construct the HJN statistic:

Step (1) Estimate θ_G from a set of N base portfolios r_t of assert returns using our proposed four-pass estimator $\widetilde{\theta}_G$

Step (2) Estimate q_G from a set of testing portfolios R_t of n assert returns and n is fixed such that $\widetilde{q}_G = \frac{1}{T}\sum_{t=1}^{T} R_t G_t'$

Step (3) The HJN statistic is

$$\widetilde{\delta}_g^2 = \left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right)' \widehat{Q}_R^{-1}\left(\iota_n - \widetilde{q}_G\widetilde{\theta}_G\right),$$

with $\widehat{Q}_R = \frac{1}{T}\sum_{t=1}^{T} R_t R_t'$.

By construction, $\sqrt{T}q_{g,T,R}\left(\widetilde{\theta}_G - \theta_G\right)$ would be of the order $O_p(\frac{1}{N^{1/2}})$ when $N/T \to c$, and thus these sampling errors would be negligible and the asymptotic distribution of the HJN statistic is determined by the distribution of $\sqrt{T}\left(\iota_n - q_{g,T,R}\theta_g\right)$ (Corollary 4.1.2).

Corollary 4.1.2. *Suppose the assumptions in Theorem 4.1 hold, then*

$$T\widetilde{\delta}_g^2 \sim_d \sum_{i=1}^{n} p_i x_i$$

with x_i *being independently* $\chi^2(1)$ *distributed random variables and* p_i *being the positive eigenvalues of the matrix* $S^{\frac{1}{2}}(\widehat{Q}_R)^{-1}S^{\frac{1}{2}'}$ *with* $S = \frac{1}{T}\sum_{t=1}^{T} e_{g,t,R}(\widetilde{\theta}_G)e_{g,t,R}(\widetilde{\theta}_G)', e_{g,t,R}(\theta_g) = \iota_N - R_t G_t' \theta_g$.

Proof: See the appendix.

Corollary 4.1.2 shows that we can make use of the HJN statistic as a valid model specification test even in the presence of weak (proxy) factors.

4.1 Simulation exercises and empirical applications

We use simulation exercises to illustrate the properties of our proposed statistics, and show their performances in the presence of weak (proxy) factors and omitted factors.

In the first simulation exercise, we calibrate to match the data set used in Anatolyev and Mikusheva (2018): the monthly returns on 100 Fama-French portfolios sorted by size and book-to-market and three Fama-French factors (g_t). From the portfolio returns we obtain the first four principal components, and we regard the first three PCs as f_t and the fourth one z_t as the omitted factors. With normalization $(f, z)'(f, z)/T = I_4$, we set the variance of these factors to be 1. We regress demeaned returns on f_t, z_t for their risk exposures, and calculate the sample mean $\mu_{\beta\gamma}$ and the sample variance $V_{\beta\gamma}$ of the risk exposures. We compute the sample variance $\sigma_e^2 I_N$ of the residuals after regressing returns on f_t, z_t. To maintain the relation between observed factors g_t and PCs f_t, we regress f_t on the three Fama French factors to obtain slope d_g and residual covariance matrix V_v. d_g would capture the quality of proxy factors.

We then simulate our data in following ways, In the first step we simulate observed factors from i.i.d $N(0, I_3)$ and latent factors f_t are generated by $Ad_g g_t + v_t$ with v_t is simulated from i.i.d $N\left(0, (I-A)d_g d_g'(I-A) + V_v\right)$. A is a diagonal matrix which we use to adjust the strengths of our (proxy) factors, and we set $A = \text{diag}(I_2, \alpha)$ in our simulations with α tuning the strength. As for the corresponding risk exposures we use $(\beta_i', \gamma_i')' \sim$ i.i.d $N(\mu_{\beta\gamma}, V_{\beta\gamma})$. Then in the end we generate $r_t = c + 1 + \beta f_t + \gamma z_t + e_t$, $e_t \sim$ $i.i.d$ $N(0, \sigma_e^2 I_N)$, where c is the sample mean of the returns from the data. In our simulations we fix $N = 100, T = 300$.

Figure 4 compares conventional estimators for θ_g, λ_g with the ones proposed in section 3. Compared with the conventional estimators, our proposed estimators are less stable. But the conventional estimators have a large drawback in 'bias', and when α is small even one-standard-deviation shadowed areas do not cover true values. Figure 5 compare the size curves of the HJ specification test and of the HJN specification test. It shows that the HJ specification test is highly size distorted even when α is large and the size distortion

increases when proxy factors become weaker (smaller α), while the HJN specification test roughly remains size correct.

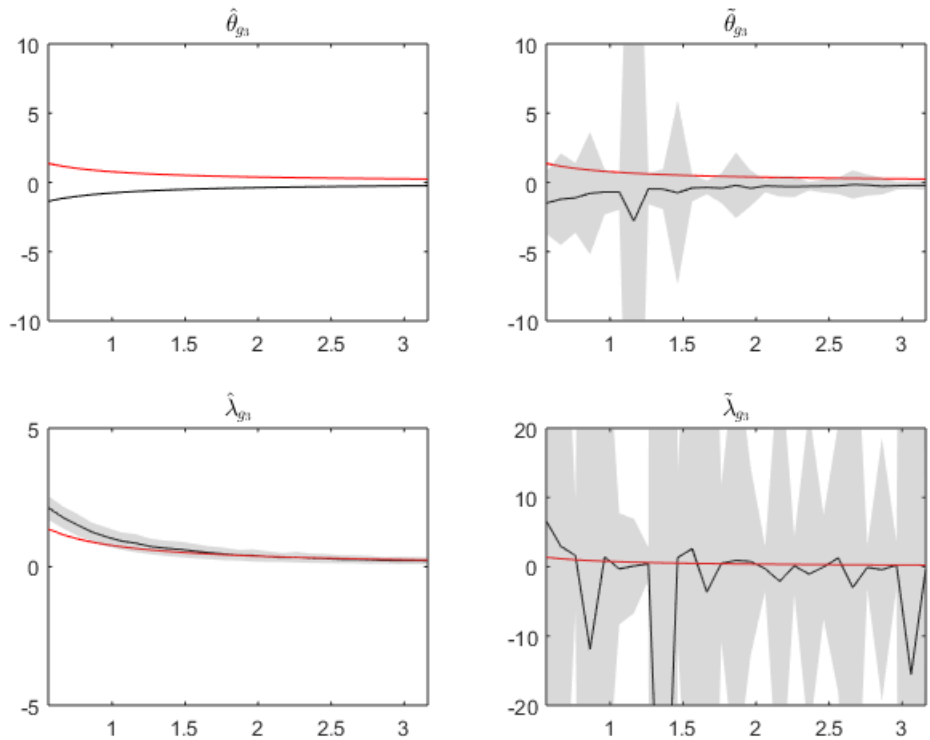

Figure 4: Sample mean (black solid curve) of the corresponding estimates with respect to the proxy factor $g_{3,t}$ against the strength of the proxy factor α, and the shadowed area covers one standard deviation from the sample mean. Red solid curves are the true values of the corresponding parameters. Left panel: conventional approach; right panel: our proposed approach in section 3.

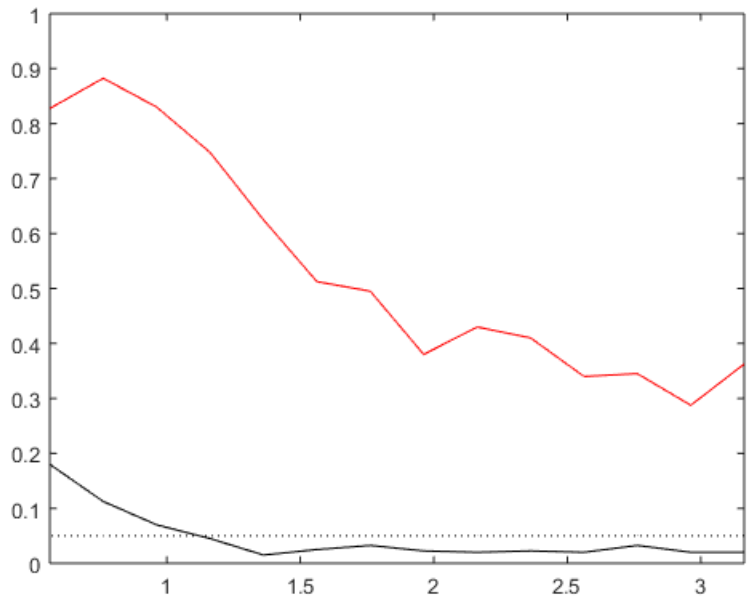

Figure 5: Rejection rates against the strength of the proxy factor α under the null hypothesis. Dotted curve: 5% line. Red solid curve: the HJ specification test. Black solid curves: the HJN specification test.

5 Empirical Application

We run our proposed procedures on the data set of the monthly returns on 100 Fama-French portfolios sorted by size and book-to-market and the three Fama-French factors (market, SmB, HmL) and the momentum factor. We see both the HJ and HJN specification tests reject the three-factor model, and while the HJ specification test rejects the four-factor model the HJN specification test does not reject it with a much larger p-value. The estimates for the four-factor model using our proposed approach indicate a negative risk premia for the momentum factor which is quite doubtful. These observations support a four-factor model for Fama French portfolios and the momentum factor might only serve as a weak (proxy) factor, which can explain the difference between the HJ and the HJN specification test results and the differences in risk premia estimations.

	Market	SMB	HML	MOM	HJ(p-val)	HJN(p-val)	HJS(rejected)
(1)					0.000	0.000	Yes
$\widehat{\theta}_G$	0.0182	-0.0313	-0.0946	–			
$\widetilde{\theta}_G$	0.2467	-0.1885	0.1258	–			
$\widehat{\lambda}_g$	0.0602	0.5279	0.3742	–			
$\widetilde{\lambda}_g$	-3.2179	0.6038	0.0347	–			
(2)					**0.001**	**0.514**	No
$\widehat{\theta}_G$	0.0248	-0.0239	-0.0592	0.1952			
$\widetilde{\theta}_G$	0.1812	0.0404	0.2959	0.8543			
$\widehat{\lambda}_g$	0.3387	0.4694	0.3730	3.1841			
$\widetilde{\lambda}_g$	-3.3109	0.3053	0.0226	-10.7839			

Table 2: Estimates with Fama-French factors and the momentum factor and tests of specification using monthly returns on 100 portfolios sorted by size and book-to-market from 1963 to 1998

6 Conclusion

We show the HJ statistic can not serve as a valid model selection tool and model specification test statistic when we take weak (proxy) factors into account. We propose weak (proxy) factors robust estimators, and two novel approaches that give size-correct model specification test results. Our empirical application supports a four factor structure for Fama French portfolios but the momentum factor may only serve as a weak proxy factor.

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A Additional assumptions and proofs

In appendix, we use $\lambda_j(A)$ to denote the jth largest eigenvalue of a given matrix A, $\lambda_{\min}(A), \lambda_{\max}(A)$ the minimum and the maximum eigenvalues. With $A = (a_{ij})$, multiple matrix norms are denoted as $\|A\| = \sqrt{\lambda_{\max}(A'A)}, \|A\|_1 = \max_j \sum_i |a_{ij}|, \|A\|_\infty = \max_i \sum_j |a_{ij}|, \|A\|_F = \sqrt{tr(A'A)}, \|A\|_{\max} = \max_{i,j} |a_{ij}|$.

A.1

Assumption A.1. *The following asymptotic distributions hold jointly:* $(\xi_{g,T}, \xi_{gg,T}, \xi_{vG,T}, \xi_{zG,T}, \xi_{eG,T},) \to_d (\xi_g, \xi_V, \xi_{vG}, \xi_{zG}, \xi_{eG})$, *where* $\xi_{g,T} = \frac{1}{\sqrt{T}}\left(\sum_{t=1}^T g_t - \mu_g\right)$, $\xi_{gg,T} = \frac{1}{\sqrt{T}}\sum_{t=1}^T \bar{g}_t\bar{g}_t'$, $\xi_{vG,T} = \frac{1}{\sqrt{T}}\sum_{t=1}^T v_t G_t'$,$\xi_{zG,T} = \frac{1}{\sqrt{T}}\sum_{t=1}^T z_t G_t'$, $\xi_{eG,T} = \frac{1}{\sqrt{T}}\sum_{t=1}^T e_t G_t'$, *and* ξ_g, $vec(\xi_{vG}), vec(\xi_{zG})$, $vec(\xi_{eG})$ *are zero-mean normal random vectors,* ξ_V *is a Gaussian random matrix.*

Assumption A.1 is a central limit theorem for the different components in equation (4) interacted with a constant and the proxy factors. Some of the statements such as $\xi_{g,T} \to_d \xi_g$ would hold if proxy factors are stationary with finite fourth moments and satisfy some strong mixing conditions (see e.g. Peligrad et al. (2006)). We specify a relatively strong assumption here instead of dealing with heavy technical details, but our results can be extended to general cases.

Proof of Lemma 2.1. Assumption 2.1 implies $r_t = c + \beta_g\bar{g}_t + \beta_g(\bar{g} - \mu_g) + \beta v_t + u_t$ then

$$\begin{aligned}\widehat{B}_g &= \sum_{i=1}^T r_i G_i' \left(\sum_{t=1}^T G_t G_t'\right)^{-1} = \sum_{i=1}^T (c + \beta_g\bar{g}_t + \beta_g(\bar{g} - \mu_g) + \beta v_t + u_t) G_i' \left(\sum_{t=1}^T G_t G_t'\right)^{-1} \\ &= (c, \beta_g) + \frac{1}{\sqrt{T}}(\beta_g\xi_{g,T}, 0) + \frac{1}{\sqrt{T}}(\beta, \gamma)\xi_{vzG,T}\widehat{Q}_G^{-1} + \frac{1}{\sqrt{T}}\xi_{eG,T}\widehat{Q}_G^{-1}\end{aligned}$$

The conclusion is then a direct result from Assumption A.1. □

Proof of Theorem 2.2. Assumption 2.1 and A.1 imply that

$$\widehat{Q}_r = Q_r + O_p(1/\sqrt{T}) \tag{15}$$

$$\iota_N = B_g Q_G \theta_G \tag{16}$$

Rewrite $T\widehat{\delta}_g^2$ as

$$\sqrt{T}\left(\iota_N - \widehat{B}_g Q_G \theta_G\right)' W \sqrt{T}\left(\iota_N - \widehat{B}_g Q_G \theta_G\right) \tag{17}$$

where

$$W = (\widehat{Q}_r^{-1} - \widehat{Q}_r^{-1}\widehat{B}_g Q_{B_g,T}(Q_{B_g,T}\widehat{B}_g'\widehat{Q}_r^{-1}\widehat{B}_g Q_{B_g,T})^{-1} Q_{B_g,T}\widehat{B}_g'\widehat{Q}_r^{-1})$$
$$Q_{B_g,T} = \text{diag}\left(I_{1+K_{g,1}}, \sqrt{T} I_{K_{g,2}}\right)$$

Lemma 2.1 and equation (16) imply that

$$\sqrt{T}\left(\iota_N - \widehat{B}_g Q_G \theta_G\right) \to_d \widetilde{\psi}_{B_g} \tag{18}$$

Lemma 2.1 and Assumption 2.3 imply the following equation holds jointly with equation (18)

$$\widehat{B}_g Q_{B_g,T} \to_d \eta_{B_g} + (0 \vdots \psi_{\beta_{g,2}}) \tag{19}$$

Plug equations (15)(18)(19) in equation (17), then we would derive the conclusion. □

Lemma A.1. *Suppose Assumption A.1 and 2.1-2.3 hold, let N,T increase and then the restrictions on e_t from Assumption 2.2 (i)(ii)(iii) also hold for $\widetilde{e}_t$ with $\widetilde{e}_t = Q_r^{-\frac{1}{2}} e_t$ with $Q_r^{-\frac{1}{2}} = A\Lambda^{-\frac{1}{2}}A'$ such that $Q_r = A\Lambda A'$ with $A'A = I_N$ and we assume $Q_r^{-\frac{1}{2}}$ is a row diagonally-dominant matrix*[1].

Proof. $\widetilde{e}_t, t = 1, \cdots, T$ are i.i.d. mean zero random vectors by construction. with finite fourth moments by construction. Next we show $\sup_i \mathbb{E}\widetilde{e}_{it}^4$ is bounded. Assumption 2.1 implies $Q_r = c'c + \beta V_f \beta' + \gamma V_z \gamma' + \Omega_e$ and thus we have the following results by eigenvalue inequalities (see e.g.

[1]In the proof of this lemma we need that $Q_r^{-\frac{1}{2}}$ has bounded absolute row sum. This is not a wild assumption if we consider the Gershgoring-type eigenvalue inclusion theorem and all eigenvalues of $Q_r^{-1/2}$ are bounded.

7.3.P16 of Horn et al. (2013)):

$$\lambda_{\max}(Q_r) = \lambda_{\max}\left(c'c + \beta V_f \beta' + \gamma V_z \gamma' + \Omega_e\right) \leq \lambda_{\max}\left(c'c + \beta V_f \beta' + \gamma V_z \gamma'\right) + L \tag{20}$$

$$\lambda_{\min}(Q_r) = \lambda_{\min}\left(c'c + \beta V_f \beta' + \gamma V_z \gamma' + \Omega_e\right) \geq \lambda_{\min}\left(c'c + \beta V_f \beta' + \gamma V_z \gamma'\right) + l = l \tag{21}$$

Then by the assumption that $Q_r^{-\frac{1}{2}}$ is a row diagonally-dominant matrix we know any row sums of $Q_r^{-\frac{1}{2}}$ would be upper bounded by $2l^{-1/2}$ and thus $\sup_i \mathbb{E}e_{it}^4 \leq L$ implies that $\sup_i \mathbb{E}\widetilde{e}_{it}^4$ is bounded. Therefore, Assumption 2.2 (i) holds.

The term $c'c + \beta V_f \beta' + \gamma V_z \gamma'$ in Q_r is a positive semi-definite matrix and we can rewrite that term as $A_{Q_r}\Lambda_{Q_r}A'_{Q_r}$ such that Λ_{Q_r} is a diagonal matrix containing all positive eigenvalues of $c'c + \beta V_f \beta' + \gamma V_z \gamma'$ and A_{Q_r} are the corresponding eigenvectors. Therefore, $Q_r = A_{Q_r}\Lambda_{Q_r}A'_{Q_r} + \Omega_e$ and thus

$$Q_r^{-1} = \Omega_e^{-1} - \Omega_e^{-1}A_{Q_r}\left(\Lambda_{Q_r}^{-1} + A'_{Q_r}\Omega_e^{-1}A_{Q_r}\right)^{-1}A'_{Q_r}\Omega_e^{-1} \tag{22}$$

which then implies that

$$\operatorname{tr}\left(\mathbb{E}Q_r^{-1/2}e_t e_t' Q_r^{-1/2}\right)/N = \operatorname{tr}\left(\Omega_e Q_r^{-1}\right)/N = 1 - \operatorname{tr}\left(\left(\Lambda_{Q_r}^{-1} + A'_{Q_r}\Omega_e^{-1}A_{Q_r}\right)^{-1}\right)/N$$

From the fact that eigenvalues of Λ_{Q_r} explode by Assumption 2.3 and eigenvalues of Ω_e are bounded by Assumption 2.2 , Courant-Fischer minimax principle implies $\operatorname{tr}\left(\left(\Lambda_{Q_r}^{-1} + A'_{Q_r}\Omega_e^{-1}A_{Q_r}\right)^{-1}\right)/N \to 0$. Furthermore, Assumption 2.2 and 2.3 imply that $\liminf_{N,T}\lambda_{\min}\left(Q_r^{-1/2}\right)$ and $\limsup_{N,T}\lambda_{\min}\left(Q_r^{-1/2}\right)$ are bounded, and thus we also have

$$0 < l < \liminf_{N,T}\lambda_{\min}\left(Q_r^{-1/2}\Omega_e Q_r^{-1/2}\right) < \limsup_{N,T}\lambda_{\max}\left(Q_r^{-1/2}\Omega_e\right)Q_r^{-1/2} < L < \infty.$$

Therefore, Assumption 2.2 (ii) holds for $\widetilde{e}_t$. As for Assumption 2.2 (iii), inequalities for the trace

of matrix product (e.g. Fang et al. (1994)) suggest

$$\begin{aligned}
&\lambda_{\min}\left(Q_r^{-1}\right)\operatorname{tr}\left(e_te_t'\right)-\lambda_{\max}\left(Q_r^{-1}\right)\operatorname{tr}\left(\mathbb{E}\left(e_te_t'\right)\right)\\
\leq&\operatorname{tr}\left(Q_r^{-1}\left(e_te_t'-\mathbb{E}\left(e_te_t'\right)\right)\right)\leq\lambda_{\max}\left(Q_r^{-1}\right)\operatorname{tr}\left(e_te_t'\right)-\lambda_{\min}\left(Q_r^{-1}\right)\operatorname{tr}\left(\mathbb{E}\left(e_te_t'\right)\right)
\end{aligned}$$

which implies that $\mathbb{E}\left|\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left(\widetilde{e}_{it}\widetilde{e}_{is}-\mathbb{E}\widetilde{e}_{it}\widetilde{e}_{is}\right)\right|^4 < L < \infty$ given all eigenvalues of Q_r^{-1} are bounded. □

Theorem A.2 illustrates how the weak (proxy) factors can affect the asymptotic properties of the estimator $\widehat{\theta}_G$. This theorem resembles theorem one in Anatolyev and Mikusheva (2018), and indeed we observe that the asymptotic behavior of the estimator $\widehat{\theta}_G$ is similar to the one of the two-stage FM risk premia in Anatolyev and Mikusheva (2018).

Theorem A.2.

Case (1): Suppose Assumption A.1 and 2.1-2.3 hold, N is fixed and T increases to infinity:

$$\sqrt{T}Q_{B_g,T}^{-1}\left(\widehat{Q}_G(\widehat{\theta}_G-\theta_G)-bias_e-bias_m\right)=O_p(1)$$

where

$$\begin{aligned}
bias_e&=-\left(\widehat{B}_g'\widehat{Q}_r^{-1}\widehat{B}_g\right)^{-1}\widehat{B}_{g,e}'\widehat{Q}_r^{-1}\widehat{B}_{g,e}\widehat{Q}_G\theta_G\\
bias_m&=-\left(\widehat{B}_g'\widehat{Q}_r^{-1}\widehat{B}_g\right)^{-1}\left(B_g+\widehat{B}_{g,m}\right)'\widehat{Q}_r^{-1}\widehat{B}_{g,m}\widehat{Q}_G\theta_G\\
\widehat{B}_{g,e}&=\frac{1}{\sqrt{T}}\xi_{eG,T}\widehat{Q}_G^{-1};\ \widehat{B}_{g,m}=\frac{1}{\sqrt{T}}\left(\beta,\gamma\right)\xi_{vzG,T}\widehat{Q}_G^{-1};\ Q_{B_g,T}=diag\left(I_{1+K_{g,1}},\sqrt{T}I_{K_{g,2}}\right)
\end{aligned}$$

and

$$\sqrt{T}Q_{B_g,T}^{-1}bias_e = \begin{pmatrix} \sqrt{T}bias_{e,1} \\ bias_{e,2} \end{pmatrix} \to_d WQ_{B_g}\widetilde{e}'Q_r^{-1}\widetilde{e}Q_G\theta_G$$

$$\sqrt{T}Q_{B_g,T}^{-1}bias_m = \begin{pmatrix} \sqrt{T}bias_{m,1} \\ bias_{m,2} \end{pmatrix} \to_d W\left(\begin{pmatrix} I_{1+K_g} \\ 0 \end{pmatrix} + \widetilde{\xi}Q_{B_g}\right)'\eta'Q_r^{-1}\eta\widetilde{\xi}Q_G\theta_G$$

$$W = -\left(\left(\eta\left(\begin{pmatrix} I_{1+K_g} \\ 0 \end{pmatrix} + \widetilde{\xi}Q_{B_g}\right) + \widetilde{e}Q_{B_g}\right)'Q_r^{-1}\left(\eta\left(\begin{pmatrix} I_{1+K_g} \\ 0 \end{pmatrix} + \widetilde{\xi}Q_{B_g}\right) + \widetilde{e}Q_{B_g}\right)\right)^{-1}$$

Case (2): Suppose Assumption A.1 and 2.1-2.3 hold, let N, T increases to infinity $N/T \to c$ and Q_r is a known row diagonally-dominant matrix such that $\eta'Q_r^{-1}\eta/N \to \widetilde{D}$:

$$\sqrt{NT}Q_{B_g,T}^{-1}\left(\widehat{Q}_G(\widehat{\theta}_G - \theta_{G,T}) - bias_e - bias_m\right) = O_p(1)$$

where

$$\theta_{G,T} = \theta_G + \widehat{Q}_G^{-1}\left(\widehat{B}_g'Q_r^{-1}\widehat{B}_g\right)^{-1}\widehat{B}_g'Q_r^{-1}\left(q_G\left(\widehat{Q}_G^{-1} - Q_G^{-1}\right) - \frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T}, 0\right)\right)\widehat{Q}_G\theta_G$$

$$\sqrt{T}Q_{B_g,T}^{-1}\widehat{Q}_G\left(\theta_{G,T} - \theta_G\right) = O_p(1)$$

$$\sqrt{T}Q_{B_g,T}^{-1}bias_e = \begin{pmatrix} \sqrt{T}bias_{e,1} \\ bias_{e,2} \end{pmatrix} \to_d WQ_{B_g}\widetilde{\Sigma}_eQ_G\theta_G$$

$$\sqrt{T}Q_{B_g,T}^{-1}bias_m = \begin{pmatrix} \sqrt{T}bias_{m,1} \\ bias_{m,2} \end{pmatrix} \to_d W\left(\begin{pmatrix} I_{1+K_g} \\ 0 \end{pmatrix} + \widetilde{\xi}Q_{B_g}\right)'\widetilde{D}\widetilde{\xi}Q_G\theta_G$$

$$W = -\left(\left(\begin{pmatrix} I_{1+K_g} \\ 0 \end{pmatrix} + \widetilde{\xi}Q_{B_g}\right)'\widetilde{D}\left(\begin{pmatrix} I_{1+K_g} \\ 0 \end{pmatrix} + \widetilde{\xi}Q_{B_g}\right) + Q_{B_g}\widetilde{\Sigma}_eQ_{B_g}\right)^{-1}$$

Proof. Case (1): After simple algebra, we can express the term $\widehat{Q}_G\left(\widehat{\theta}_G - \theta\right) - bias_e - bias_m$ in the

following way

$$\widehat{Q}_G\left(\widehat{\theta}_G-\theta\right)-bias_e-bias_m=\left(\widehat{B}_g'\widehat{Q}_r^{-1}\widehat{B}_g\right)^{-1}\left\{\widehat{B}_g'\widehat{Q}_r^{-1}\left(q_G\left(\widehat{Q}_G^{-1}-Q_G^{-1}\right)-\frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T},0\right)\right)\right.$$
$$\left.-\left(\frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T},0\right)+\widehat{B}_{g,e}\right)'\widehat{Q}_r^{-1}\widehat{B}_{g,m}-\left(B_g+\frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T},0\right)+\widehat{B}_{g,m}\right)'\widehat{Q}_r^{-1}\widehat{B}_{g,e}\right\}\widehat{Q}_G\theta_G \tag{23}$$

From the proofs of Lemma 2.1 and Theorem 2.2 we have

$$\widehat{B}_g=B_g+\frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T},0\right)+\widehat{B}_{g,m}+\widehat{B}_{g,e} \tag{24}$$
$$\widehat{B}_gQ_{B_g,T}=\left(c+O_p(1/\sqrt{T}),\eta_{\beta_g}\right)+\left(\frac{1}{\sqrt{T}}\widetilde{e}_T+\frac{1}{\sqrt{T}}\eta\widetilde{\xi}_T\right)Q_{B_g,T} \tag{25}$$

where $\widetilde{\xi}_T=\sqrt{T}\widehat{B}_{g,m}=\begin{pmatrix}0 & 0\\ \left(d_gQ_{\beta_g}\right)^{-1} & 0\\ 0 & I_{K_z}\end{pmatrix}\xi_{vzG,T}\widehat{Q}_G^{-1},\ \ \widetilde{e}_T=\sqrt{T}\widehat{B}_{g,e}=\xi_{eG,T}\widehat{Q}_G^{-1},\widetilde{\xi}_T\to_d\widetilde{\xi},\widetilde{e}_T\to_d\widetilde{e},Q_{B_g,T}/\sqrt{T}\to Q_{B_g}=\mathrm{diag}\left(0_{1+K_{g,1}},I_{K_{g,2}}\right)$ and $\widetilde{\xi},\widetilde{e}$ are Gaussian random matrices. Equation (25) implies that

$$\widehat{B}_gQ_{B_g,T}\to_d\ \eta\left(\begin{pmatrix}I_{1+K_g}\\ 0\end{pmatrix}+\widetilde{\xi}Q_{B_g}\right)+\widetilde{e}Q_{B_g} \tag{26}$$

With the above intermediate results, we prove our statement with respect to the term $bias_e$ and the rests follow the same steps. Assumption A.1 implies that the above asymptotic results hold jointly and thus if we plug these into the equation below we would derive the asymptotic distribution of the term $bias_e$

$$\sqrt{T}Q_{B_g,T}^{-1}bias_e=-\left((\widehat{B}_gQ_{B_g,T})'\widehat{Q}_r^{-1}(\widehat{B}_gQ_{B_g,T})\right)^{-1}(\widehat{B}_{g,e}Q_{B_g,T})'\widehat{Q}_r^{-1}(\sqrt{T}\widehat{B}_{g,e})\widehat{Q}_G\theta_G \tag{27}$$

Case (2): Next we discuss the case where N,T both grows to infinity. We first show the following

two results:

$$(2.1)\quad T\widetilde{e}_T' Q_r^{-1}\widetilde{e}_T/N \to_p \tilde{\Sigma}_e \tag{28}$$

$$(2.2)\quad \sqrt{T}\widetilde{e}_T' Q_r^{-1}\eta/N \to_p 0 \tag{29}$$

To prove the statement (2.1), let $\widetilde{e}_t = Q_r^{-\frac{1}{2}} e_t$ and $\rho(s,t) = \frac{1}{\sqrt{N}}\sum_{i=1}^N \widetilde{e}_{it}'\widetilde{e}_{is}$. By construction, $\mathbb{E}\rho^2(s,t) = \frac{1}{N}\text{tr}\left(\Omega_{\widetilde{e}}\Omega_{\widetilde{e}}\right) \leq \lambda_{\max}^2(\Omega_{\widetilde{e}})$ with $\Omega_{\widetilde{e}} = \mathbb{E}\widetilde{e}_t\widetilde{e}_t'$ and thus Lemma A.1 implies that $\mathbb{E}\rho^2(s,t)$ is bounded. Assumption 2.1 (finite fourth moments of proxy factors), Assumption 2.2 (i) and the bounded $\mathbb{E}\rho^2(s,t)$ deliver the following inequality

$$\begin{aligned}
&\mathbb{E}\left(\frac{1}{T\sqrt{N}}\sum_{i=1}^N\left(\sum_{t=1}^T\sum_{s\neq t}^T G_{tm}G_{sn}\widetilde{e}_{it}'\widetilde{e}_{is}\right)\right)^2\\
=&\mathbb{E}\left(\frac{1}{T}\left(\sum_{t=1}^T\sum_{s\neq t}^T G_{tm}G_{sn}\rho(s,t)\right)\right)\left(\frac{1}{T}\left(\sum_{t'=1}^T\sum_{s'\neq t}^T G_{t'm}G_{s'n}\rho(s',t')\right)\right)\\
=&\mathbb{E}\left(\frac{1}{T^2}\left(\sum_{t=1}^T\sum_{s\neq t}^T\left(G_{tm}^2G_{sn}^2 + G_{tm}G_{sn}G_{sm}G_{tn}\right)\rho^2(s,t)\right)\right) < L
\end{aligned} \tag{30}$$

which via Chebyshev's inequality gives

$$\frac{1}{T\sqrt{N}}\sum_{i=1}^N\left(\sum_{t=1}^T\sum_{s\neq t}^T G_tG_s'\widetilde{e}_{it}'\widetilde{e}_{is}\right) = O_P(1) \tag{31}$$

Finally we arrive at (2.1):

$$T\widetilde{e}_T' Q_r\widetilde{e}_T/N = \widehat{Q}_g^{-1}\left(\frac{1}{NT}\sum_{t=1}^T G_t\widetilde{e}_t'\widetilde{e}_tG_t'\right)\widehat{Q}_g^{-1} + O_p(1/\sqrt{N}) = \tilde{\Sigma}_e + O_p(1/\sqrt{N}) \tag{32}$$

where the first equality is due to equation (31) and the last equality is guaranteed by Lemma A.1.

Next, we prove the statement (2.2). We first look at the second moments

$$\mathbb{E}\left(\left\|\frac{1}{\sqrt{T}}\sum_{t=1}^{T}G_te_t'Q_r^{-1}\eta/\sqrt{N}\right\|^2\right)\leq\mathbb{E}\left(\operatorname{tr}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T}G_t\widetilde{e}_t'Q_r^{-1/2}\eta/\sqrt{N}\right)\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T}G_t\widetilde{e}_t'Q_r^{-1/2}\eta/\sqrt{N}\right)'\right)$$

$$=\frac{1}{T}\sum_{t=1}^{T}\operatorname{tr}\left(\mathbb{E}\left(G_t\widetilde{e}_t'Q_r^{-1/2}\eta\eta'Q_r^{-1/2}\widetilde{e}_tG_t'\right)/N\right)\leq L\operatorname{tr}\left(\Omega_{\widetilde{e}}Q_r^{-1/2}\eta\eta'Q_r^{-1/2}/N\right)\leq L\lambda_{\max}\left(\Omega_{\widetilde{e}}\right)\left\|Q_r^{-1/2}\right\|_1^2\left(\|\eta\|_F^2/N\right)$$

where the first inequality is because factors g_t have finite fourth moments. The proof of Lemma A.1 shows that $\lambda_{\max}\left(\Omega_{\widetilde{e}}\right)$ and $\left\|Q_r^{-1/2}\right\|_1^2$ are bounded, Assumption 2.3 implies that $\|\eta\|_F^2/N$ is bounded. Therefore, we know

$$\mathbb{E}\left(\left\|\frac{1}{\sqrt{T}}\sum_{t=1}^{T}G_te_t'Q_r^{-1}\eta/\sqrt{N}\right\|^2\right)\leq L<\infty \tag{33}$$

and thus (2.2) holds since $\sqrt{T}\widetilde{e}_T'Q_r^{-1}\eta/N=\frac{1}{\sqrt{N}}Q_g^{-1}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T}G_te_t'Q_r^{-1}\eta/\sqrt{N}\right)=O_p(1/\sqrt{N})$. In the end, $\eta'Q_r^{-1}\eta/N\to\widetilde{D}$ and the result (2.2) imply the following term is of order $O_p(1)$ when $N/T\to c$:

$$\frac{\sqrt{T}}{\sqrt{N}}Q_{B_g,T}\left\{-\left(\frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T},0\right)+\frac{1}{\sqrt{T}}\widetilde{e}_T\right)'Q_r^{-1}\frac{1}{\sqrt{T}}\widetilde{\xi}_T-\left(B_g+\frac{1}{\sqrt{T}}\left(\beta_g\xi_{g,T},0\right)+\frac{1}{\sqrt{T}}\widetilde{\xi}_T\right)'Q_r^{-1}\frac{1}{\sqrt{T}}\widetilde{e}_T\right\}=O_p(1)$$

□

Assumption A.2. *Let $e_{g,t}=\iota_N-r_tG_t'\theta_G$, and the restrictions on e_t from Assumption 2.2 (i)(ii)(iii) also hold for $e_{g,t}$.*

Proof of Corollary 2.2.2. Theorem 2.2 suggest for given N, $T\widehat{\delta}_g^2\to_d d_{1,N}$ with $d_{1,N}=\widetilde{\psi}_{B_g}'M_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g}+(0\vdots\psi_{\beta_{g,2}})\right)}\widetilde{\psi}_{B_g}$, $\widetilde{\psi}_{B_g}\sim N(0,S_{B_g})$ and $S_{B_g}=p\lim S_{B_g,T}$, $S_{B_g,T}=p\lim\frac{1}{T}\sum_{t=1}^{T}e_{g,t}e_{g,t}'$. From the construction of the $\widehat{c}_{1-\alpha}$, we know it is drawn from the distribution of the $d_{2,N}=\psi_S'M_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g}+(0\vdots\psi_{\beta_{g,2}})\right)}\psi_S$ with $\psi_S\sim N(0,S)$, $S=p\lim S_T$, $S_T=\frac{1}{T}\sum_{t=1}^{T}\widehat{e}_{g,t}\widehat{e}_{g,t}'$, $\widehat{e}_{g,t}=\iota_N-r_tG_t'\widehat{\theta}_G$ and ψ_S independent from ψ_{B_g}.

The proof of Theorem A.2 suggest that $\widetilde{\psi}_{B_g}'P_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g}+(0\vdots\psi_{\beta_{g,2}})\right)}\widetilde{\psi}_{B_g}=O_p(1)$ for any given N,

and the same for the term $\psi_S' P_{Q_r^{-\frac{1}{2}}\left(\eta_{B_g}+(0\vdots\psi_{\beta_{g,2}})\right)}\psi_S$. Now we look at the difference $S_{B_g,T} - S_T$.

$$
\begin{aligned}
S_T =& \frac{1}{T}\sum_{t=1}^{T}\left(\iota_N - r_t G_t'\left(\widehat{\theta}_G - \theta_G + \theta_G\right)\right)\left(\iota_N - r_t G_t'\left(\widehat{\theta}_G - \theta_G + \theta_G\right)\right)' \\
=& S_{B_g,T} + \frac{1}{T}\sum_{t=1}^{T}\left(r_t G_t'\left(\widehat{\theta}_G - \theta_G\right)\right)\left(r_t G_t'\left(\widehat{\theta}_G - \theta_G\right)\right)' \\
& - \frac{1}{T}\sum_{t=1}^{T}\left(r_t G_t'\left(\widehat{\theta}_G - \theta_G\right)\right)e_{g,t}' - \frac{1}{T}\sum_{t=1}^{T} e_{g,t}\left(r_t G_t'\left(\widehat{\theta}_G - \theta_G\right)\right)' \qquad (34)
\end{aligned}
$$

Assumption A.2 and proofs from Theorem A.2 then suggest the last two terms be negligible and $S_T \geq S_{B_g,T} + A + o_p(1)$ in large samples with $tr(A)$ of order $O_p(N)$. This then lead to the conclusion.

□

A.2 Consistent θ estimator

Assumption A.3. $\|(\beta;\gamma)_i\| = \|c_{\beta\gamma,i}\| \leq L$. $Q_{(\beta;\gamma)} = \lim_{N\to\infty}(\beta;\gamma)'(\beta;\gamma)/N$ *with* $Q_{(\beta;\gamma)}$ *a* $K_{vz}\times K_{vz}$ *positive definite matrix with* $0 < l \leq \lambda_{K_g}\left(Q_{(\beta;\gamma)}\right) \leq \lambda_1\left(Q_{(\beta;\gamma)}\right) < L < \infty$. *(Assumption 2.3 implies this assumption via the Ostrowski theorem and some extra mild assumptions on* Q_r*.) such that*

$$\left\|N^{-1}c_{\beta\gamma}'c_{\beta\gamma} - Q_{\beta\gamma}\right\| = o_p(1)$$

Assumption A.4. *Let* $\gamma_N(t,t') = \mathbb{E}\left(N^{-1}\sum_{i=1}^{N} e_{it}e_{it'}\right)$, *there exists a positive constant* L *such that*

$$(1) T^{-1}\sum_{t=1}^{T}\sum_{t'=1}^{T}|\gamma_N(t,t')| \leq L; \max_t |\gamma_N(t,t')| \leq L$$

$$(2) T^{-2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\mathbb{E}\left(\sum_{i=1}^{N} e_{it}e_{it'} - \mathbb{E}\left(\sum_{i=1}^{N} e_{it}e_{it'}\right)\right)^2 = T^{-2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\left(\mathbb{E}\left(\sum_{i=1}^{N} e_{it}e_{it'}\right)^2 - N^2\gamma_N(t,t')^2\right) \leq LN$$

Assumption A.4 is implies by Assumption 2.2.

Assumption A.5. $\mathbb{E}\left|\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left(e_{it}e_{is} - \mathbb{E}e_{it}e_{is}\right)\right|^4 < L < \infty$

Assumption A.6. *This is implied by previous assumptions again. (Weak dependence between proxy factors and Idiosyncratic Errors from Bai and Ng (2002))*

$$(i) \quad \mathbb{E}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{\sqrt{T}}\sum_{t=1}^{T} g_t e_{it}\right\|^2\right) \leq L$$

$$(ii) \quad \mathbb{E}\left(\frac{1}{K_{vz}}\sum_{i=1}^{K_{vz}}\left\|\frac{1}{\sqrt{T}}\sum_{t=1}^{T} g_t u_{vz,it}\right\|^2\right) \leq L$$

$$(iii) \quad \mathbb{E}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\frac{1}{\sqrt{T}}\sum_{t=1}^{T} u_{vz,t} e_{it}\right\|^2\right) \leq L$$

Assumption A.7. *For all N and T,*

$$\mathbb{E}\sum_{t=1}^{T}\left(\sum_{i=1}^{N} c_{\beta\gamma,ij} e_{it}\right)^2 \leq LNT$$

$$\mathbb{E}\left(\sum_{t=1}^{T}\sum_{i=1}^{N} c_{\beta\gamma,ij} e_{it}\right)^2 \leq LNT \tag{35}$$

Assumption A.8.

$$\left\|T^{-1}u_{vz}'u_{vz} - \Sigma_{vz}\right\|_{\max} = O_p(T^{-\frac{1}{2}})$$

with Σ_{vz} a positive definite matrix.

Assumption A.9.

$$\sup_t \mathbb{E}\left\|u_{vz,t}\right\|^4 \leq L$$

This is one assumption identical to one imposed on factors (Assumption A) in Bai and Ng (2002) and Anatolyev and Mikusheva (2018) impose slightly stronger assumption.

Assumption A.10.

$$(1) \sum_{t}^{T} |\gamma_N(s,t)| \leq L$$
$$(2) \sum_{i=1}^{N} |\tau(i,j)| \leq L$$

with $\mathbb{E}e_{it}e_{jt} = \tau_t(i,j), |\tau_t(i,j)| \leq |\tau(i,j)|$.

Assumption A.11. *(i)*

$$\frac{1}{NT}\mathbb{E}\left\|\sum_{s=1}^{T}\sum_{n=1}^{N} u_{vz,s}\left(e_{is}e_{it} - \mathbb{E}(e_{is}e_{it})\right)\right\|^2 \leq L$$

(ii) for each t, as $N \to \infty$

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N} c_{\beta\gamma,i}e_{ti} \to_d N(0,\Pi_t)$$

with $\Pi_t = \lim_{N\to\infty} \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} c_{\beta\gamma,i}c'_{\beta\gamma,j}\mathbb{E}e_{ti}e_{tj}$. *And for all* $j = 1, \cdots, N$

$$\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T} c_{\beta\gamma,i}\left(e_{ti}e_{tj} - \mathbb{E}e_{ti}e_{tj}\right) = O_p(1)$$

(iii)

$$\mathbb{E}\left\|\frac{1}{\sqrt{NT}}\sum_{t=1}^{T}\sum_{i=1}^{N} u_{vz,t}c'_{\beta\gamma,i}e_{it}\right\|_F^2 \leq M$$

Assumption A.11 is identical to the Assumption F in Bai (2003).

This section mainly contains the following steps: (1) we first show that the number of the strong factors can be estimated consistently; (2) we then show common component can be estimated consistently when $\sqrt{N}/T \to 0$, and we also provide a consistent estimator for Q_r; (3.0) we show that with proxies for the common component, θ can be consistently derived even in the presence of weak identification issues (here we still assume $\sqrt{N}/T \to 0$) by regressing on both $\widehat{\beta}_g$ and the common component estimator; (3.1) we also derive the asymptotic distribution; (4) compare with

the other approaches; (5) in stage 3 we do not split the sample, and in this stage we split sample into testing and base portfolios and apply the results from step 3.

A.3 Step (1)

In this section we prove that we can estimate the number of the strong factors in the $u_{g,t}$ consistently. Here we only provide one way to estimate the number, the estimation approach is not unique. Bai and Ng (2002) propose multiple consistent estimators for the number of strong factors with different penalty functions. Here we use the one employed in Giglio and Xiu (2017).

Under Assumption 2.1, we know

$$r_t = \widetilde{c} + \beta d_g \bar{g}_t + u_{g,t} \tag{36}$$

with $u_{g,t} = \beta v_t + \gamma z_t + e_t, \widetilde{c} = c + \beta d_g(\bar{g} - \mu_g)$. v_t is assumed to be of a $K_g \times 1$ vector, but the dimension of z_t is unknown. We estimate the number K_{vz} of the omitted strong factors by

$$\widehat{K}_{vz} = \arg \min_{K_g \leq j \leq K_{vz,\max}} \left(N^{-1}T^{-1}\lambda_j\left(\widehat{u}_g\widehat{u}_g'\right) + j\phi(N,T)\right) - 1 \tag{37}$$

where $\widehat{u}_g$ is $T \times N$ matrix stacked with the residuals $\widehat{u}_{g,t}$, $K_{vz,\max}$ is an arbitrary upper bound for K_{vz} and $\phi(N,T)$ is a penalty function with the properties $\phi(N,T) \to 0, \phi(N,T)/(N^{-\frac{1}{2}} + T^{-\frac{1}{2}}) \to \infty$. Now we show this estimator is consistent.

Proof. We basically follow the steps in Giglio and Xiu (2017) with small changes in the middle.

(1) We first prove the claim such that for $1 \leq j \leq K_{vz}$

$$|N^{-1}T^{-1}\lambda_j\left(\widehat{u}_g\widehat{u}_g'\right) - \lambda_j\left((Q_{(\beta;\gamma)})^{\frac{1}{2}}\Sigma_{(v;z)}(Q_{(\beta;\gamma)})^{\frac{1}{2}}\right)| = o_p(1) \tag{38}$$

with $Q_{(\beta;\gamma)} = \lim_{N\to\infty}(\beta;\gamma)'(\beta;\gamma)/N$.

For convenience, in this proof, denote $c_{\beta\gamma} = (\beta;\gamma), u_{vz} = (v;z)$.

(1.1) Notice

$$\widehat{u}_g\widehat{u}_g' - M_{\bar{G}}u_{vz}c_{\beta\gamma}'c_{\beta\gamma}u_{vz}'M_{\bar{G}} = M_{\bar{G}}u_{vz}c_{\beta\gamma}'e'M_{\bar{G}} + M_{\bar{G}}ec_{\beta\gamma}u_{vz}'M_{\bar{G}} + M_{\bar{G}}ee'M_{\bar{G}}$$

with $\widehat{u}_g = M_{\bar{G}} r$. We show in the following steps that the three terms on the right is negligible when divided by NT.

For the term $M_{\bar{G}} ee' M_{\bar{G}}$, we have

$$\left\| M_{\bar{G}} ee' M_{\bar{G}} - n\Gamma_u \right\| \leq \left\| ee' - n\Gamma_u \right\|_F + 2 \left\| P_{\bar{G}} ee' \right\|_F + \left\| P_{\bar{G}} ee' P_{\bar{G}} \right\|_F \tag{39}$$

above (1)

$$\mathbb{E} \left\| e \right\|_F^2 = \mathbb{E} \sum_{i=1}^{N} \sum_{t=1}^{T} e_{it}^2 \leq LNT \tag{40}$$

$$\mathbb{E} \left\| \bar{e} \right\|_F^2 = T^{-2} \mathbb{E} \sum_{i=1}^{N} \sum_{t=1}^{T} \sum_{t'=1}^{T} e_{it} e_{it'} = NT^{-2} \sum_{t=1}^{T} \sum_{t'=1}^{T} |\gamma_N(t, t')| \leq LNT^{-1} \tag{41}$$

above (2)

$$\mathbb{E} \left\| ee' - n\Gamma_u \right\|_F^2 = \sum_{t=1}^{T} \sum_{t'=1}^{T} \mathbb{E} \left(\sum_{i=1}^{N} e_{it} e_{it'} - \mathbb{E} \left(\sum_{i=1}^{N} e_{it} e_{it'} \right) \right)^2 \leq LNT^2 \tag{42}$$

$$\mathbb{E} \left\| G' e \right\|_F^2 = NT \mathbb{E} \left(\frac{1}{N} \sum_{i=1}^{N} \left\| \frac{1}{\sqrt{T}} \sum_{t=1}^{T} G_t e_{it} \right\|_F^2 \right) \leq LNT \tag{43}$$

$$\left\| G' e \right\|_F = O_p(N^{\frac{1}{2}} T^{\frac{1}{2}}); \left\| \left(0, \iota_T \bar{g}' \right)' e \right\|_F = O_p(N^{\frac{1}{2}} T^{\frac{1}{2}}) \tag{44}$$

$$\left\| \bar{G}' e \right\|_F \leq \left\| G' e \right\|_F + \left\| \left(0, \iota_T \bar{g}' \right)' e \right\|_F = O_p(N^{\frac{1}{2}} T^{\frac{1}{2}}) \tag{45}$$

$$\|P_{\bar{G}}e\|_F \leq T^{-1} \left\|\bar{G}\right\|_F \left\|\left(\bar{G}'\bar{G}/T\right)^{-1}\right\|_F \left\|\bar{G}'e\right\|_F = O_p(N^{\frac{1}{2}}) \tag{46}$$

Then it follows that

$$\begin{aligned}\left\|M_{\bar{G}}ee'M_{\bar{G}} - n\Gamma_u\right\| \leq& \left\|ee' - n\Gamma_u\right\|_F + 2\left\|P_{\bar{G}}ee'\right\|_F + \left\|P_{\bar{G}}ee'P_{\bar{G}}\right\|_F \\ =& O_p(N^{\frac{1}{2}}T) + O_p(NT^{\frac{1}{2}}) + O_p(N)\end{aligned} \tag{47}$$

From assumption A.6 (1) we would have

$$\mathbb{E}\left\|\Gamma_u\right\|_F^2 = \mathbb{E}\sum_{t=1}^{T}\sum_{t'=1}^{T}|\gamma_N(t,t')| \leq LT \tag{48}$$

$$\left\|M_{\bar{G}}ee'M_{\bar{G}}\right\|_F \leq \left\|M_{\bar{G}}ee'M_{\bar{G}} - n\Gamma_u\right\|_F + \left\|n\Gamma_u\right\|_F = O_p(N^{\frac{1}{2}}T) + O_p(NT^{\frac{1}{2}})$$

For the term $M_{\bar{G}}u_{vz}c'_{\beta\gamma}e'M_{\bar{G}}$, we have

$$\begin{aligned}\left\|M_{\bar{G}}u_{vz}c'_{\beta\gamma}e'M_{\bar{G}}\right\|_F \leq& \left\|u_{vz}c'_{\beta\gamma}e'\right\|_F + \left\|P_{\bar{G}}u_{vz}c'_{\beta\gamma}e'\right\|_F + \left\|u_{vz}c'_{\beta\gamma}e'P_{\bar{G}}\right\|_F + \left\|P_{\bar{G}}u_{vz}c'_{\beta\gamma}e'P_{\bar{G}}\right\|_F \\ \leq& \left(2(K_{vz}+1)\right)^2\left\|u_{vz}c'_{\beta\gamma}e'\right\|_F\end{aligned} \tag{49}$$

The above assumption is not from previous assumptions, and they can be derived from the assumptions imposed in Anatolyev and Mikusheva (2018).

$$\mathbb{E}\left\|ec_{\beta\gamma}\right\|_F^2 = \mathbb{E}\sum_{j=1}^{K_{vz}}\sum_{t=1}^{T}\left(\sum_{i=1}^{N}c_{\beta\gamma,ij}e_{it}\right)^2 \leq LNT \tag{50}$$

Thus

$$\|u_{vz}\| \leq KT\left\|T^{-1}u'_{vz}u_{vz}\right\|_{\max} = O_p(T^{\frac{1}{2}}) \tag{51}$$

$$\left\|u_{vz}c'_{\beta\gamma}e'\right\| \leq \|u_{vz}\| \left\|c'_{\beta\gamma}e'\right\|_F = O_p(N^{\frac{1}{2}}T) \tag{52}$$

$$\left\|M_{\bar{G}}u_{vz}c'_{\beta\gamma}e'M_{\bar{G}}\right\| \leq O_p(N^{\frac{1}{2}}T) \tag{53}$$

Therefore

$$N^{-1}T^{-1}|\lambda_j(\widehat{u}_g\widehat{u}'_g) - \lambda_j(M_{\bar{G}}u_{vz}c'_{\beta\gamma}c_{\beta\gamma}u'_{vz}M_{\bar{G}})| = o_p(1) \tag{54}$$

(1.2) To finish the proof of part (1) we only need to show the following two results:

$$\text{(1.2.1)} \quad |N^{-1}T^{-1}\lambda_j(M_{\bar{G}}u_{vz}c'_{\beta\gamma}c_{\beta\gamma}u'_{vz}M_{\bar{G}}) - T^{-1}\lambda_j(M_{\bar{G}}u_{vz}Q_{\beta\gamma}u'_{vz}M_{\bar{G}})| = o_p(1) \tag{55}$$

$$\text{(1.2.2)} \quad |T^{-1}\lambda_j(M_{\bar{G}}u_{vz}Q_{\beta\gamma}u'_{vz}M_{\bar{G}}) - \lambda_j\left((Q_{(\beta;\gamma)})^{\frac{1}{2}}\Sigma_{(v;z)}(Q_{(\beta;\gamma)})^{\frac{1}{2}}\right)| = o_p(1) \tag{56}$$

The equation 55 is a direct result of Assumption A.3 and Weyl's inequality such that

$$\begin{aligned}&|N^{-1}T^{-1}\lambda_j(M_{\bar{G}}u_{vz}c'_{\beta\gamma}c_{\beta\gamma}u'_{vz}M_{\bar{G}}) - T^{-1}\lambda_j(M_{\bar{G}}u_{vz}Q_{\beta\gamma}u'_{vz}M_{\bar{G}})| \\ \leq & T^{-1}\left\|N^{-1}c'_{\beta\gamma}c_{\beta\gamma} - Q_{\beta\gamma}\right\| \|M_{\bar{G}}u_{vz}\|^2 = o_p(1)\end{aligned} \tag{57}$$

The equation 55 is a direct result of Assumption A.8 and Weyl's inequality such that

$$\begin{aligned}&|T^{-1}\lambda_j(M_{\bar{G}}u_{vz}Q_{\beta\gamma}u'_{vz}M_{\bar{G}}) - \lambda_j\left((Q_{(\beta;\gamma)})^{\frac{1}{2}}\Sigma_{vz}(Q_{(\beta;\gamma)})^{\frac{1}{2}}\right)| \\ \leq & L\left\|T^{-1}u'_{vz}u_{vz} - \Sigma_{vz}\right\| = O_p(T^{-\frac{1}{2}})\end{aligned} \tag{58}$$

Therefore, for $1 \leq j \leq K_{vz}$

$$|N^{-1}T^{-1}\lambda_j\left(\widehat{u}_g\widehat{u}'_g\right) - \lambda_j\left((Q_{(\beta;\gamma)})^{\frac{1}{2}}\Sigma_{(v;z)}(Q_{(\beta;\gamma)})^{\frac{1}{2}}\right)| = o_p(1) \tag{59}$$

(2) In part (2) we finish the proof of the consistency $\widehat{K}_{vz}$ by showing the following statement

$$\lim_{T,N\to\infty}\inf \mathbb{P}\left(\widehat{F}(K_{vz}+1)\leq \widehat{F}(j), j=1,\cdots,N\right)=1 \tag{60}$$

with $\widehat{F}(j)=N^{-1}T^{-1}\lambda_j\left(\widehat{u}_g\widehat{u}_g'\right)+j\phi(N,T)$. Notice a direct result from step (1) is that for $1\leq j\leq K_{vz}$, we can find $l>0$ such that $\widehat{F}(j)>l+o_p(1)$, while for $K_{vz}+1\leq j$, $\widehat{F}(j)=o_p(1)$. Then we only need to show that for $j>K_{vz}+1$, $\widehat{F}(j)>\widehat{F}(K_{vz}+1)$ with probability approaching to one. Notice for $j\geq K_{vz}+1$,

$$\lambda_j\left(\widehat{u}_g\widehat{u}_g'\right)\leq \left\|ee'\right\|=O_p(TN^{-\frac{1}{2}}+T^{-\frac{1}{2}}N) \tag{61}$$

and this implies $\widehat{F}(j)>\widehat{F}(K_{vz}+1)$ with probability approaching to one as $(N^{\frac{1}{2}}T^{\frac{1}{2}})(j-K_{vz}-1)\phi(N,T)>(N^{-\frac{1}{2}}T^{-\frac{1}{2}})\left(\lambda_{K_{vz}+1}\left(\widehat{u}_g\widehat{u}_g'\right)-\lambda_j\left(\widehat{u}_g\widehat{u}_g'\right)\right)$ with probability approaching to one.

□

A.4 Step (2)

Denote $m_{NT}=\min\{N,T\}$.

Lemma A.3.

$$m_{NT}\frac{1}{T}\sum_{t=1}^{T}\left\|\widetilde{u}_{vz,t}-H'u_{vz,t}\right\|_F^2=O_p(1)$$

with $H=\left(c_{\beta\gamma}'c_{\beta\gamma}/N\right)\left(u_{vz}'\widetilde{u}_{vz}/T\right)V_{NT}$ *and* V_{NT} *being the* $K_{vz}\times K_{vz}$ *diagonal matrix of* $\lambda_i\left(\widehat{u}_g\widehat{u}_g'/NT\right), i=1,\cdots,K_{vz}$.

Proof. This proof resembles the proof of Theorem 1 in Bai and Ng (2002). From the normalization $\widetilde{u}_{vz}'\widetilde{u}_{vz}/T=I_{K_{vz}}$ we know

$$\|\widetilde{u}_{vz}\|_F^2=\sum_{t=1}^{T}\|\widetilde{u}_{vz,t}\|_F^2=O_p(T) \tag{62}$$

From the proof in section B.1 and the above equation we know

$$\|H\|_F \leq \left\|c'_{\beta\gamma}c_{\beta\gamma}/N\right\|_F \left\|u'_{vz}u_{vz}/T\right\|_F^{1/2} \left\|\widetilde{u}'_{vz}\widetilde{u}_{vz}/T\right\|_F^{1/2} = O_p(1) \tag{63}$$

From the equation (78) and (38), we know

$$\left\|\widetilde{u}_{vz,t} - H'u_{vz,t}\right\|_F^2 \leq L\sum_{i=1}^{6} a_{i,t}^2 \tag{64}$$

with

$$\begin{aligned}
a_{1,t}^2 =& T^{-2}\left\|\sum_{s=1}^{T}\widetilde{u}_{vz,s}\gamma_N(s,t)\right\|_F^2 \\
a_{2,t}^2 =& T^{-2}\left\|\sum_{s=1}^{T}\widetilde{u}_{vz,s}\zeta_{ee,st}\right\|_F^2 \\
a_{3,t}^2 =& T^{-2}\left\|\sum_{s=1}^{T}\widetilde{u}_{vz,s}\zeta_{ue,st}\right\|_F^2 \\
a_{4,t}^2 =& T^{-2}\left\|\sum_{s=1}^{T}\widetilde{u}_{vz,s}\zeta_{eu,st}\right\|_F^2 \\
a_{5,t}^2 =& \left\|\frac{1}{NT}\widetilde{u}'_{vz}P_{\bar{G}}u_g u'_{g,t}\right\|_F^2 \\
a_{6,t}^2 =& \left\|\frac{1}{NT}\widetilde{u}'_{vz}M_{\bar{G}}u_g\left(\widehat{u}_{g,t} - u_{g,t}\right)\right\|_F^2
\end{aligned} \tag{65}$$

From the equation (62) and Assumption A.4 (i),

$$\sum_{t=1}^{T} a_{1,t}^2 \leq \left(T^{-1}\sum_{s=1}^{T}\|\widetilde{u}_{vz,s}\|_F^2\right)\left(T^{-1}\sum_{t=1}^{T}\sum_{s=1}^{T}\gamma_N(s,t)^2\right) = O_p(1) \tag{66}$$

From the equation (62) and Assumption 2.2,

$$\sum_{t=1}^{T} a_{2,t}^2 \leq \left(T^{-1}\sum_{s=1}^{T}\|\widetilde{u}_{vz,s}\|_F^2\right)\left(T^{-2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\sum_{u=1}^{T}\zeta_{ee,su}\zeta_{ee,ut}\right)^2\right)^{\frac{1}{2}} = O_p(T/N) \tag{67}$$

and from the equation (50)

$$\sum_{t=1}^{T} a_{3,t}^2 \leq \left(\sum_{t=1}^{T} \left\|c_{\beta\gamma}' e_t/N\right\|_F^2\right) \left(T^{-1}\sum_{s=1}^{T} \|\widetilde{u}_{vz,s}\|_F^2\right) \left(T^{-1}\sum_{s=1}^{T} \|u_{vz,s}\|_F^2\right) = O_p(T/N) \tag{68}$$

and similarly $\sum_{t=1}^{T} a_{4,t}^2 \leq O_p(T/N)$.

From Assumption 2.2,

$$\|P_{\bar{G}} u_g\|_F \leq T^{-1} \left\|\bar{G}\right\|_F \left\|\left(\bar{G}'\bar{G}/T\right)^{-1}\right\|_F \left\|\bar{G}' u_g\right\|_F = O_p(N^{\frac{1}{2}}) \tag{69}$$

and thus

$$\sum_{t=1}^{T} a_{5,t}^2 \leq \left(T^{-1}\sum_{s=1}^{T} \|\widetilde{u}_{vz,s}\|_F^2\right) \left(N^{-1} \|P_{\bar{G}} u_g\|_F^2\right) \left(N^{-1}T^{-1}\sum_{t=1}^{T} \|u_{g,t}\|_F^2\right) = O_p(1) \tag{70}$$

From the equation (38) and (61)

$$\|M_{\bar{G}} u_g\|_F^2 = O_p(NT) \tag{71}$$

and since $\sqrt{T}\left(\widehat{\beta}_g - \beta_g\right) = O_p(1)$

$$\sum_{t=1}^{T} \|\widehat{u}_{g,t} - u_{g,t}\|_F^2 \leq \left\|\sqrt{T}\left(\widehat{\beta}_g - \beta_g\right)\right\|_F^2 \left\|\bar{G}/\sqrt{T}\right\|_F^2 \leq O_p(N) \tag{72}$$

Therefore,

$$\sum_{t=1}^{T} a_{6,t}^2 \leq \left(T^{-1}\sum_{s=1}^{T} \|\widetilde{u}_{vz,s}\|_F^2\right) \left(N^{-1}T^{-1} \|M_{\bar{G}} u_g\|_F^2\right) \left(N^{-1}\sum_{t=1}^{T} \|\widehat{u}_{g,t} - u_{g,t}\|_F^2\right) \leq O_p(1) \tag{73}$$

It follows

$$\frac{1}{T}\sum_{t=1}^{T} \left\|\widetilde{u}_{vz,t} - H' u_{vz,t}\right\|_F^2 = O_p(1/N) + O_p(1/T) \tag{74}$$

□

Given the discussion in the last subsection, it would be safe to assume that we know the value

of K_{vz}, namely the number of the strong omitted factors are known. In this section, we show the asymptotic properties of the common component estimator. Essentially, this section is a building block for the consistent θ_g estimator as introduced in the next section.

The method of principal components method gives the following estimators

$$(\widetilde{c}_{\beta\gamma}, \widetilde{u}_{vz}) = \arg \min_{c_i, u_t \ s.t. \ \sum_{t=1}^{T} u_t u_t'/T = I_{K_{vz}}} \sum_{i,t} \left(\widehat{u}_{g,it} - c_i' u_t\right)^2 \tag{75}$$

The estimator $\widetilde{u}_{vz}$ is equal to the $\sqrt{T}$ times eigenvector associated with the K_{vz} largest eigenvalues of the matrix $\widehat{u}_g \widehat{u}_g'$, and $\widetilde{c}_{\beta\gamma} = \widetilde{u}_{vz}' \widehat{u}_g / T$ corresponds to the OLS estimator regressing $\widehat{u}_g$ over $\widetilde{u}_v z$. Specially we are interested in the common component estimator $\widetilde{c} = \widetilde{u}_{vz} (\widetilde{c}_{\beta\gamma})'$, which would serve as a proxy for the common component in u_g in our proposed estimator.

Theorem A.4. *Assumption ? hold, and when* $d\sqrt{N}/T \to 0$,

$$\sqrt{N}\left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right) = V_{NT}^{-1} \frac{1}{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} u_{vz,s}'\right) \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_{\beta\gamma,i} e_{it} + o_p(1) \tag{76}$$

when $\liminf \sqrt{N}/T \to \tau > 0$,

$$T\left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right) = O_p(1) \tag{77}$$

Proof. We make use the equation $\widetilde{u}_{vz} = \frac{1}{NT} M_{\bar{G}} u_g u_g' M_{\bar{G}} \widetilde{u}_{vz} V_{NT}^{-1}$, and $H = \left(c_{\beta\gamma}' c_{\beta\gamma}/N\right)\left(u_{vz}' \widetilde{u}_{vz}/T\right) V_{NT}^{-1}$, and V_{NT} is the $K_{vz} \times K_{vz}$ diagonal matrix of $\lambda_i\left(\widehat{u}_g \widehat{u}_g'/NT\right), i = 1, \cdots, K_{vz}$. Then

$$\begin{aligned}
&\widetilde{u}_{vz,t} - H' u_{vz,t} \\
=&\frac{1}{NT} V_{NT}^{-1} \widetilde{u}_{vz}' \left(M_{\bar{G}} u_g \widehat{u}_{g,t} - u_{vz}\left(c_{\beta\gamma}' c_{\beta\gamma}\right) u_{vz,t}\right) \\
=&\frac{1}{NT} V_{NT}^{-1} \widetilde{u}_{vz}' \left(u_g u_{g,t} - u_{vz}\left(c_{\beta\gamma}' c_{\beta\gamma}\right) u_{vz,t} - P_{\bar{G}} u_g u_{g,t} + M_{\bar{G}} u_g \left(\widehat{u}_{g,t} - u_{g,t}\right)\right) \\
=&\frac{1}{NT} V_{NT}^{-1} \widetilde{u}_{vz}' \left(\left(u_{vz} c_{\beta\gamma}' + e\right)\left(c_{\beta\gamma} u_{vz,t} + e_t\right) - u_{vz}\left(c_{\beta\gamma}' c_{\beta\gamma}\right) u_{vz,t} - P_{\bar{G}} u_g u_{g,t} + M_{\bar{G}} u_g \left(\widehat{u}_{g,t} - u_{g,t}\right)\right) \\
=&V_{NT}^{-1}\left(\frac{1}{T}\sum_{s=1}^{T} \widetilde{u}_{vz,s} \gamma_N(s,t) + \frac{1}{T}\sum_{s=1}^{T} \widetilde{u}_{vz,s} \zeta_{ee,st} + \frac{1}{T}\sum_{s=1}^{T} \widetilde{u}_{vz,s} \zeta_{ue,st} + \frac{1}{T}\sum_{s=1}^{T} \widetilde{u}_{vz,s} \zeta_{eu,st}\right) \\
&+ V_{NT}^{-1}\left(-\frac{1}{NT} \widetilde{u}_{vz}' P_{\bar{G}} u_g u_{g,t} + \frac{1}{NT} \widetilde{u}_{vz}' M_{\bar{G}} u_g \left(\widehat{u}_{g,t} - u_{g,t}\right)\right)
\end{aligned} \tag{78}$$

with

$$\zeta_{ee,st} = e_s' e_t / N - \gamma_N(s,t) \tag{79}$$

$$\zeta_{ue,st} = u_{vz,s}' c_{\beta\gamma}' e_t / N \tag{80}$$

$$\zeta_{eu,st} = u_{vz,t}' c_{\beta\gamma}' e_s / N \tag{81}$$

Now we analyze the terms on the left hand of equation (78).

(1) Assumption A.9 and Assumption A.4 imply that

$$\frac{1}{T} \sum_{s=1}^{T} u_{vz,s} \gamma_N(s,t) = O_p(1/T) \tag{82}$$

Lemma A.3 and Assumption A.4 imply that

$$\begin{aligned} \frac{1}{T} \left\| \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} - u_{vz,s} \right) \gamma_N(s,t) \right\|_F &\leq \frac{1}{\sqrt{T}} \left(\frac{1}{T} \sum_{s=1}^{T} \| \widetilde{u}_{vz,s} - u_{vz,s} \|_F^2 \right)^{1/2} \left(\sum_{t}^{T} |\gamma_N(s,t)|^2 \right)^{1/2} \\ &= O_p(1/(\sqrt{T m_{NT}})) \end{aligned} \tag{83}$$

and thus

$$\frac{1}{T} \sum_{s=1}^{T} \widetilde{u}_{vz,s} \gamma_N(s,t) = \frac{1}{T} \sum_{s=1}^{T} u_{vz,s} \gamma_N(s,t) + \frac{1}{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} - u_{vz,s} \right) \gamma_N(s,t) \leq O_p(1/(\sqrt{T m_{NT}})) \tag{84}$$

(2)

Assumption A.11 implies that

$$\frac{1}{T} \sum_{s=1}^{T} u_{vz,s} \zeta_{ee,st} = O_p(1/\sqrt{NT}) \tag{85}$$

Assumption 2.2 implies

$$\frac{1}{T} \sum_{s=1}^{T} |\zeta_{ee,st}|^2 = \frac{1}{T} \frac{1}{N} \sum_{s=1}^{T} \left| \frac{1}{\sqrt{N}} \left(e_{is} e_{it} - \mathbb{E}(e_{is} e_{it}) \right) \right|^2 = O_p(1/(N)) \tag{86}$$

Lemma A.3 and equation (86) imply that

$$\frac{1}{T}\left\|\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-u_{vz,s}\right)\zeta_{ee,st}\right\| \leq \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-u_{vz,s}\right\|^2\right)^{1/2}\left(\frac{1}{T}\sum_{s=1}^{T}\left|\zeta_{ee,st}\right|^2\right)^{1/2}$$
$$=O_p(1/(\sqrt{Nm_{NT}})) \tag{87}$$

and thus

$$\frac{1}{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}\zeta_{ee,st}=\frac{1}{T}\sum_{s=1}^{T}u_{vz,s}\zeta_{ee,st}+\frac{1}{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-u_{vz,s}\right)\zeta_{ee,st}\leq O_p(1/(\sqrt{Nm_{NT}})) \tag{88}$$

(3) Assumption A.11 implies that

$$\frac{1}{T}\sum_{s=1}^{T}u_{vz,s}\zeta_{ue,st}=\left(\frac{1}{T}\sum_{s=1}^{T}u_{vz,s}u'_{vz,s}\right)\frac{1}{\sqrt{N}}\left(\frac{1}{\sqrt{N}}\sum_{i=1}^{N}c_{\beta\gamma,i}e_{ti}\right)=O_p(1/\sqrt{N}) \tag{89}$$

and

$$\frac{1}{T}\sum_{s=1}^{T}\left|\zeta_{ue,st}\right|^2=\frac{1}{T}\sum_{s=1}^{T}\left(u'_{vz,s}c'_{\beta\gamma}e_t/N\right)^2\leq\left\|u_{vz}\right\|_F/T\left\|c'_{\beta\gamma}e'_t/\sqrt{N}\right\|_F/N=O_p(1/N) \tag{90}$$

Lemma A.3 and equation (90) imply that

$$\frac{1}{T}\left\|\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-u_{vz,s}\right)\zeta_{ue,st}\right\| \leq \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-u_{vz,s}\right\|^2\right)^{1/2}\left(\frac{1}{T}\sum_{t}^{T}\left|\zeta_{ue,st}\right|^2\right)^{1/2}$$
$$=O_p(1/(\sqrt{Nm_{NT}})) \tag{91}$$

and thus

$$\frac{1}{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}\zeta_{ue,st}=\frac{1}{T}\sum_{s=1}^{T}u_{vz,s}\zeta_{ue,st}+\frac{1}{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-u_{vz,s}\right)\zeta_{ue,st}=O_p(1/\sqrt{N}) \tag{92}$$

(4) Lemma A.3 and Assumption A.11 imply that

$$\left\|\frac{1}{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-u_{vz,s}\right)e_s'c_{\beta\gamma}/Nu_{vz,t}\right\|_F \leq \frac{1}{\sqrt{N}}\left(\frac{1}{T}\sum_{s=1}^{T}\|\widetilde{u}_{vz,s}-u_{vz,s}\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{s=1}^{T}\left\|e_s'c_{\beta\gamma}/\sqrt{N}\right\|_F^2\right)^{\frac{1}{2}}\|u_{vz,t}\|_F$$
$$=O_p(1/\sqrt{Nm_{NT}}) \tag{93}$$

Assumption A.11 implies

$$\frac{1}{\sqrt{NT}}\left(\frac{1}{\sqrt{NT}}\sum_{s=1}^{T}u_{vz,s}e_s'c_{\beta\gamma}\right)u_{vz,t}=O_p(\frac{1}{\sqrt{NT}}) \tag{94}$$

and thus

$$\frac{1}{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}\zeta_{ue,st}=\frac{1}{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-u_{vz,s}\right)e_s'c_{\beta\gamma}/Nu_{vz,t}+\frac{1}{T}\sum_{s=1}^{T}u_{vz,s}e_s'c_{\beta\gamma}/Nu_{vz,t}\leq O_p(1/\sqrt{Nm_{NT}}) \tag{95}$$

(5) Similar to the derivation of equation (45)

$$\left\|\bar{G}'u_g\right\|_F=O_p(N^{\frac{1}{2}}T^{\frac{1}{2}});\quad \left\|\bar{G}'u_{vz}\right\|_F=O_p(T^{\frac{1}{2}}) \tag{96}$$

which implies

$$\begin{aligned}&\frac{1}{NT}\left\|\left(\widetilde{u}_{vz}-u_{vz}\right)'\bar{G}\left(\bar{G}'\bar{G}\right)^{-1}\bar{G}'u_gu_{g,t}'\right\|_F\\ \leq&\left(\frac{1}{T}\sum_{t=1}^{T}\|\widetilde{u}_{vz,t}-u_{vz,t}\|_F^2\right)^{\frac{1}{2}}\frac{1}{\sqrt{T}}\left(\left\|\bar{G}/\sqrt{T}\left(\bar{G}'\bar{G}/T\right)^{-1}\right\|_F\right)\left(\frac{1}{\sqrt{NT}}\left\|\bar{G}'u_g\right\|_F\right)\left\|u_{g,t}/\sqrt{N}\right\|_F\\ =&O_p(1/\sqrt{Tm_{NT}})\end{aligned} \tag{97}$$

and

$$
\begin{aligned}
&\frac{1}{NT}\left\|u_{vz}'\bar{G}\left(\bar{G}'\bar{G}\right)^{-1}\bar{G}'u_g u_{g,t}'\right\|_F \\
\leq&\left(\frac{1}{\sqrt{T}}\left\|u_{vz}'\bar{G}\right\|_F\right)\frac{1}{T}\left(\left\|\left(\bar{G}'\bar{G}/T\right)^{-1}\right\|_F\right)\left(\frac{1}{\sqrt{NT}}\left\|\bar{G}'u_g\right\|_F\right)\left\|u_{g,t}/\sqrt{N}\right\|_F \\
=&O_p(1/T)
\end{aligned}
\tag{98}
$$

Therefore,

$$
\begin{aligned}
\frac{1}{NT}\widetilde{u}_{vz}'P_{\bar{G}}u_g u_{g,t}' =&\frac{1}{NT}\left(\widetilde{u}_{vz}-u_{vz}\right)'\bar{G}\left(\bar{G}'\bar{G}\right)^{-1}\bar{G}'u_g u_{g,t}'+\frac{1}{NT}u_{vz}'\bar{G}\left(\bar{G}'\bar{G}\right)^{-1}\bar{G}'u_g u_{g,t}' \\
\leq&O_p(1/\sqrt{Tm_{NT}})
\end{aligned}
\tag{99}
$$

(6)

$$
\frac{1}{NT}\left\|u_{vz}'u_{vz}c_{\beta\gamma}'\left(\widehat{u}_{g,t}-u_{g,t}\right)\right\|_F \leq\frac{1}{\sqrt{NT}}\left\|u_{vz}'u_{vz}/T\right\|_F\left\|\frac{\sqrt{T}c_{\beta\gamma}'\left(\widehat{u}_{g,t}-u_{g,t}\right)}{\sqrt{N}}\right\|_F=O_p(\frac{1}{\sqrt{NT}})
\tag{100}
$$

$$
\frac{1}{NT}\left\|u_{vz}'e\left(\widehat{u}_{g,t}-u_{g,t}\right)\right\|_F\leq\frac{1}{T}\left\|u_{vz}'e/\sqrt{NT}\right\|_F\left\|\frac{\sqrt{T}\left(\widehat{u}_{g,t}-u_{g,t}\right)}{\sqrt{N}}\right\|_F=O_p(\frac{1}{T})
\tag{101}
$$

and thus

$$
\frac{1}{NT}\left\|\widetilde{u}_{vz}'M_{\bar{G}}u_g\right\|_F\leq\frac{1}{NT}\left\|\widetilde{u}_{vz}'u_g\left(\widehat{u}_{g,t}-u_{g,t}\right)\right\|_F+\frac{1}{NT}\left\|\widetilde{u}_{vz}'P_{\bar{G}}u_g\left(\widehat{u}_{g,t}-u_{g,t}\right)\right\|_F=O_p(\frac{1}{\sqrt{NT}})+O_p(\frac{1}{T})
\tag{102}
$$

From the above discussion, when $\sqrt{N}/T\to 0$, only the third term in equation 78 matters in the asymptotic behavior and thus

$$
\sqrt{N}\left(\widetilde{u}_{vz,t}-H'u_{vz,t}\right)=V_{NT}^{-1}\frac{1}{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}u_{vz,s}'\right)\frac{1}{\sqrt{N}}\sum_{i=1}^{N}c_{\beta\gamma,i}e_{it}+o_p(1)
\tag{103}
$$

when $\liminf \sqrt{N}/T \to \tau > 0$, it is straightforward that

$$T\left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right) = O_p(1) \tag{104}$$

□

Theorem A.5. *(factor loading)*$\sqrt{T}/N \to 0$

$$\sqrt{T}\left(\widetilde{c}_{\beta\gamma,i} - H^{-1} c_{\beta\gamma,i}\right) = H' \frac{1}{\sqrt{T}} u_{vz}' e_i + o_p(1)$$

Proof.

$$\begin{aligned}
\widetilde{c}_{\beta\gamma,i} =& \frac{1}{T} \widetilde{u}_{vz}' \widehat{u}_{g,i} \\
=& \frac{1}{T} \widetilde{u}_{vz}' \left(\left(\widetilde{u}_{vz} H^{-1} + u_{vz} - \widetilde{u}_{vz} H^{-1}\right) c_{\beta\gamma,i} + e_i\right) - \frac{1}{T} \widetilde{u}_{vz}' \left(\widehat{u}_{g,i} - u_{g,i}\right) \\
=& H^{-1} c_{\beta\gamma,i} + \frac{1}{T} H' u_{vz}' e_i + \frac{1}{T} \widetilde{u}_{vz}' \left(u_{vz} - \widetilde{u}_{vz} H^{-1}\right) c_{\beta\gamma,i} + \frac{1}{T}\left(\widetilde{u}_{vz}' - H' u_{vz}'\right) e_i - \frac{1}{T} \widetilde{u}_{vz}' \left(\widehat{u}_{g,i} - u_{g,i}\right)
\end{aligned} \tag{105}$$

We show the last three terms are at most $O_p(1/m_{NT})$.

(1) For the term $\frac{1}{T}\left(\widetilde{u}_{vz} - u_{vz} H\right)' e_i$, equation 78 implies

$$\begin{aligned}
&\frac{1}{T}\left(\widetilde{u}_{vz} - u_{vz} H\right)' e_i = V_{NT}^{-1} \left(\frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \widetilde{u}_{vz,s} \gamma_N(s,t) e_{it} + \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \widetilde{u}_{vz,s} \zeta_{ee,st} e_{it} + \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \widetilde{u}_{vz,s} \zeta_{ue,st} e_{it} \right) \\
&+ V_{NT}^{-1} \left(\frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \widetilde{u}_{vz,s} \zeta_{eu,st} e_{it} - \frac{1}{NT^2} \sum_{t=1}^{T} \widetilde{u}_{vz}' P_{\bar{G}} u_g u_{g,t} e_{it} + \frac{1}{NT^2} \sum_{t=1}^{T} \widetilde{u}_{vz}' M_{\bar{G}} u_g \left(\widehat{u}_{g,t} - u_{g,t}\right) e_{it} \right) \\
&= V_{NT}^{-1} \sum_{j=1}^{6} a_j
\end{aligned} \tag{106}$$

There are six terms on the left hand side of the above equation, and we analyze each of $a_j, j = 1, \cdots, 6$ to determine the order of $\frac{1}{T}\left(\widetilde{u}_{vz} - u_{vz} H\right)' e_i$.

Assumption 2.2 and Lemma 1(i) in Bai and Ng (2002) imply that

$$\left\| \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} - H' u_{vz,s} \right) \gamma_N(s,t) e_{it} \right\|_F$$
$$\leq \frac{1}{\sqrt{T}} \left(\frac{1}{T} \sum_{s=1}^{T} \left\| \widetilde{u}_{vz,s} - H' u_{vz,s} \right\|_F^2 \right)^{\frac{1}{2}} \left(\frac{1}{T} \sum_{t=1}^{T} \sum_{s=1}^{T} \gamma_N(s,t)^2 \frac{1}{T} \sum_{t=1}^{T} e_{it}^2 \right)^{\frac{1}{2}} \leq O_p(\frac{1}{\sqrt{T m_{NT}}}) \tag{107}$$

Assumption 2.2,A.9 and Lemma 1(i) in Bai and Ng (2002) imply that

$$\mathbb{E} \left\| \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} u_{vz,s} \gamma_N(s,t) e_{it} \right\|_F \leq \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \left(\mathbb{E} \left\| u_{vz,s} \right\|_F^2 \right)^{\frac{1}{2}} \gamma_N(s,t)^2 (\mathbb{E} e_{it}^2)^{\frac{1}{2}} = O(\frac{1}{T}) \tag{108}$$

Therefore,

$$a_1 = \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} - H' u_{vz,s} \right) \gamma_N(s,t) e_{it} + \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} H' u_{vz,s} \gamma_N(s,t) e_{it} \leq O_p(\frac{1}{\sqrt{T m_{NT}}}) \tag{109}$$

Assumption 2.2 implies that

$$\frac{1}{T} \sum_{t=1}^{T} \zeta_{ee,st} e_{it} = \frac{1}{\sqrt{N}} \frac{1}{T} \sum_{t=1}^{T} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^{N} \left(e_{is} e_{it} - \mathbb{E} \left(e_{is} e_{it} \right) \right) \right) e_{it} = O_p(\frac{1}{\sqrt{N}}) \tag{110}$$

Equation (110) and Lemma A.3 imply that

$$\left\| \frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} - H' u_{vz,s} \right) \zeta_{ee,st} e_{it} \right\|_F \leq \left(\frac{1}{T} \sum_{s=1}^{T} \left\| \widetilde{u}_{vz,s} - H' u_{vz,s} \right\|_F^2 \right)^{\frac{1}{2}} \left(\frac{1}{T} \sum_{s=1}^{T} \left(\frac{1}{T} \sum_{t=1}^{T} \zeta_{ee,st} e_{it} \right)^2 \right)^{\frac{1}{2}}$$
$$\leq O_p(\frac{1}{\sqrt{N m_{NT}}}) \tag{111}$$

Assumption A.11 implies that

$$\frac{1}{T^2} \sum_{t=1}^{T} \sum_{s=1}^{T} u_{vz,s} \zeta_{ee,st} e_{it} = \frac{1}{\sqrt{NT}} \left(\frac{1}{T} \sum_{t=1}^{T} \left(\frac{1}{\sqrt{NT}} \sum_{s=1}^{T} \sum_{i=1}^{N} u_{vz,s} \left(e_{is} e_{it} - \mathbb{E} \left(e_{is} e_{it} \right) \right) \right) e_{it} \right)$$
$$\leq O_p(\frac{1}{\sqrt{NT}}) \tag{112}$$

and thus

$$a_2 = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s} - H'u_{vz,s}\right)\zeta_{ee,st}e_{it} + \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}H'u_{vz,s}\zeta_{ee,st}e_{it} \leq O_p(\frac{1}{\sqrt{Nm_{NT}}}) \tag{113}$$

Assumption A.11 implies that

$$\frac{1}{T}\sum_{t=1}^{T}\zeta_{ue,st}e_{it} = \frac{1}{\sqrt{N}}u'_{vz,s}\left(\frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{\sqrt{N}}\sum_{i=1}c_{\beta\gamma,i}e_{it}\right)e_{it}\right) = O_p(\frac{1}{\sqrt{N}}) \tag{114}$$

and thus with Theorem A.4 we know

$$\begin{aligned}\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s} - H'u_{vz,s}\right)\zeta_{ue,st}e_{it}\right\|_F \leq& \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s} - H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{s=1}^{T}\left(\frac{1}{T}\sum_{t=1}^{T}\zeta_{ue,st}e_{it}\right)^2\right)^{\frac{1}{2}}\\ \leq& O_p(\frac{1}{\sqrt{Nm_{NT}}})\end{aligned} \tag{115}$$

Assumption A.10 and A.3 imply that

$$\frac{1}{NT}\sum_{t=1}^{T}\sum_{j=1}^{N}\left\|c_{\beta\gamma,i}\tau_t(i,j)\right\|_F \leq L\frac{1}{N}\sum_{j=1}^{N}\left\|\tau(i,j)\right\|_F = O_p(\frac{1}{N}) \tag{116}$$

Assumption A.11 implies that $\frac{1}{NT}\sum_{t=1}^{T}\sum_{j=1}^{N}c_{\beta\gamma,i}\left(e_{jt}e_{it} - \tau_t(i,j)\right) = O_p(\frac{1}{\sqrt{NT}})$ and thus with equation (116) we know

$$\begin{aligned}\frac{1}{NT}\sum_{t=1}^{T}\sum_{j=1}^{N}c_{\beta\gamma,i}e_{jt}e_{it} =& \frac{1}{NT}\sum_{t=1}^{T}\sum_{j=1}^{N}c_{\beta\gamma,i}\left(e_{jt}e_{it} - \tau_t(i,j) + \tau_t(i,j)\right)\\ \leq& O_p(\frac{1}{\sqrt{NT}}) + O_p(\frac{1}{N})\end{aligned} \tag{117}$$

Therefore,

$$a_3 = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s} - H'u_{vz,s}\right)\zeta_{ue,st}e_{it} + \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}H'u_{vz,s}\zeta_{ue,st}e_{it} \leq O_p(\frac{1}{m_{NT}}) \tag{118}$$

Assumption A.11 implies that

$$\frac{1}{T}\sum_{t=1}^{T}\zeta_{eu,st}e_{it} = \frac{1}{\sqrt{N}}\left(\frac{1}{T}\sum_{t=1}^{T}e_{it}u'_{vz,t}\left(\frac{1}{\sqrt{N}}\sum_{j=1}^{N}c_{\beta\gamma,j}e_{js}\right)\right) = O_p(\frac{1}{\sqrt{N}}) \tag{119}$$

and thus with Theorem A.4 we know

$$\begin{aligned}\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-H'u_{vz,s}\right)\zeta_{eu,st}e_{it}\right\|_F &\leq \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{s=1}^{T}\left(\frac{1}{T}\sum_{t=1}^{T}\zeta_{ue,st}e_{it}\right)^2\right)^{\frac{1}{2}}\\ &\leq O_p(\frac{1}{\sqrt{Nm_{NT}}})\end{aligned} \tag{120}$$

Assumption A.11 implies that

$$\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}u_{vz,s}\zeta_{eu,st}e_{it} = \frac{1}{\sqrt{NT}}\left(\frac{1}{\sqrt{NT}}\sum_{s=1}^{T}u_{vz,s}e'_s c_{\beta\gamma}\right)\left(\frac{1}{T}\sum_{t=1}^{T}u_{vz,t}e_{it}\right) = O_p(\frac{1}{\sqrt{NT}}) \tag{121}$$

Therefore,

$$a_4 = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-H'u_{vz,s}\right)\zeta_{eu,st}e_{it} + \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}H'u_{vz,s}\zeta_{eu,st}e_{it} \leq O_p(\frac{1}{m_{NT}}) \tag{122}$$

Similar to the derivation of equation (98)

$$\left\|\frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}P_{\bar{G}}u_g u_{g,t}e_{it}\right\|_F \leq \left\|\frac{1}{NT}\sum_{t=1}^{T}\widetilde{u}'_{vz}P_{\bar{G}}u_g\right\|_F\frac{\sqrt{N}}{\sqrt{T}}\left\|\frac{1}{\sqrt{NT}}\sum_{t=1}^{T}u_{g,t}e_{it}\right\|_F = O_p(\frac{1}{T}) \tag{123}$$

$$a_5 = -\frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}P_{\bar{G}}u_g u_{g,t}e_{it} \leq O_p(\frac{1}{T}) \tag{124}$$

Equation (102) implies that

$$a_6 = \frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}M_{\bar{G}}u_g\left(\widehat{u}_{g,t}-u_{g,t}\right)e_{it} \leq O_p(\frac{1}{m_{NT}}) \tag{125}$$

(2) For the term $\frac{1}{T}\widetilde{u}'_{vz}\left(u_{vz}-\widetilde{u}_{vz}H^{-1}\right)c_{\beta\gamma,i}$

Similar to the analysis of the term $\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)' e_i$, equation 78 implies

$$
\begin{aligned}
&\frac{1}{T}\sum_{t=1}^{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)' u_{vz} \\
=&V_{NT}^{-1}\left(\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}u'_{vz,t}\gamma_N(s,t)+\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}u'_{vz,t}\zeta_{ee,st}+\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}u'_{vz,t}\zeta_{ue,st}\right) \\
&+V_{NT}^{-1}\left(\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\widetilde{u}_{vz,s}u'_{vz,t}\zeta_{eu,st}-\frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}P_{\bar{G}}u_g u_{g,t}u'_{vz,t}+\frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}M_{\bar{G}}u_g\left(\widehat{u}_{g,t}-u_{g,t}\right)u'_{vz,t}\right) \\
&=V_{NT}^{-1}\sum_{j=1}^{6}a_j
\end{aligned}
\tag{126}
$$

Following the same proof as the one for a_1 in the term $\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)' e_i$, Assumption A.8 and Lemma 1(i) in Bai and Ng (2002) imply that

$$
\begin{aligned}
&\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-H'u_{vz,s}\right)\gamma_N(s,t)u'_{vz,t}\right\|_F \\
\leq&\frac{1}{\sqrt{T}}\left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}\gamma_N(s,t)^2\frac{1}{T}\sum_{t=1}^{T}u'_{vz,t}u_{vz,t}\right)^{\frac{1}{2}}\leq O_p(\frac{1}{\sqrt{Tm_{NT}}})
\end{aligned}
\tag{127}
$$

and Assumption A.9 and Lemma 1(i) in Bai and Ng (2002) imply that

$$
\mathbb{E}\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}u_{vz,s}\gamma_N(s,t)u_{vz,t}\right\|_F\leq\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\mathbb{E}\left\|u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\gamma_N(s,t)^2(\mathbb{E}\left\|u_{vz,t}\right\|_F^2)^{\frac{1}{2}}=O(\frac{1}{T})
\tag{128}
$$

and thus

$$
a_1\leq O_p(\frac{1}{\sqrt{Tm_{NT}}})
\tag{129}
$$

Assumption A.11 and Lemma A.3 imply that

$$\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-H'u_{vz,s}\right)\zeta_{ee,st}u'_{vz,t}\right\|_F \leq \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{s=1}^{T}\left\|\frac{1}{T}\sum_{t=1}^{T}\zeta_{ee,st}u'_{vz,t}\right\|_F^2\right)^{\frac{1}{2}}$$
$$\leq O_p(\frac{1}{\sqrt{NTm_{NT}}}) \tag{130}$$

Assumption A.11 implies

$$\frac{1}{T^2}\sum_{s=1}^{T}\sum_{t=1}^{T}u_{vz,s}u'_{vz,t}\zeta_{ee,st}=\frac{1}{\sqrt{NT}}\left(\frac{1}{T}\sum_{s=1}^{T}u_{vz,s}\left(\frac{1}{\sqrt{NT}}\sum_{t=1}^{T}\sum_{i=1}^{N}\left(e_{it}e_{is}-\mathbb{E}e_{it}e_{is}\right)u'_{vz,t}\right)\right)=O_p(\frac{1}{\sqrt{NT}}) \tag{131}$$

and thus

$$a_2=\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-H'u_{vz,s}\right)u'_{vz,t}\zeta_{ee,st}+\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}H'u_{vz,s}u'_{vz,t}\zeta_{ee,st}\leq O_p(\frac{1}{m_{NT}}) \tag{132}$$

Assumption A.11 implies that

$$\frac{1}{T}\sum_{t=1}^{T}\zeta_{ue,st}u'_{vz,t}=\frac{1}{\sqrt{NT}}u'_{vz,s}\left(\frac{1}{\sqrt{NT}}\sum_{t=1}^{T}\sum_{i=1}^{N}c_{\beta\gamma,i}e_{it}u'_{vz,t}\right)=O_p(\frac{1}{\sqrt{NT}}) \tag{133}$$

and thus

$$\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}u_{vz,s}\zeta_{ue,st}u'_{vz,t}=\frac{1}{\sqrt{NT}}\frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}u_{vz,s}u'_{vz,s}\left(\frac{1}{\sqrt{NT}}\sum_{t=1}^{T}\sum_{i=1}^{N}c_{\beta\gamma,i}e_{it}u'_{vz,t}\right)=O_p(\frac{1}{\sqrt{NT}}) \tag{134}$$

and with Lemma A.3 we know

$$\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s}-H'u_{vz,s}\right)\zeta_{ue,st}u_{vz,t}\right\|_F \leq \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{s=1}^{T}\left(\frac{1}{T}\sum_{t=1}^{T}\zeta_{ue,st}u_{vz,t}\right)^2\right)^{\frac{1}{2}}$$
$$\leq O_p(\frac{1}{\sqrt{NTm_{NT}}}) \tag{135}$$

Therefore,

$$a_3 = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s} - H'u_{vz,s}\right)\zeta_{ue,st}u'_{vz,t} + \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}H'u_{vz,s}\zeta_{ue,st}u'_{vz,t} \leq O_p(\frac{1}{\sqrt{NT}}) \quad (136)$$

Assumption A.11 implies that

$$\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}u_{vz,s}\zeta_{eu,st}u'_{vz,t} = \frac{1}{\sqrt{NT}}\left(\frac{1}{\sqrt{NT}}\sum_{s=1}^{T}\sum_{i=1}^{N}u_{vz,s}c'_{\beta\gamma,i}e_{is}\right)\left(\frac{1}{T}\sum_{t=1}^{T}u_{vz,t}u'_{vz,t}\right) = O_p(\frac{1}{\sqrt{NT}}) \quad (137)$$

and

$$\frac{1}{T}\sum_{t=1}^{T}\zeta_{eu,st}u_{vz,t} = \frac{1}{\sqrt{N}}\left(\frac{1}{\sqrt{N}}\sum_{i=1}^{N}c'_{\beta\gamma,i}e_{is}\right)\left(\frac{1}{T}\sum_{t=1}^{T}u_{vz,t}u'_{vz,t}\right) = O_p(\frac{1}{\sqrt{N}}) \quad (138)$$

and thus with Lemma A.3 we know

$$\begin{aligned}\left\|\frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s} - H'u_{vz,s}\right)\zeta_{ue,st}u_{vz,t}\right\|_F &\leq \left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s} - H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\left(\frac{1}{T}\sum_{s=1}^{T}\left(\frac{1}{T}\sum_{t=1}^{T}\zeta_{ue,st}u_{vz,t}\right)^2\right)^{\frac{1}{2}} \\ &\leq O_p(\frac{1}{\sqrt{Nm_{NT}}}) \end{aligned} \quad (139)$$

$$a_4 = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\left(\widetilde{u}_{vz,s} - H'u_{vz,s}\right)\zeta_{eu,st}u'_{vz,t} + \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}H'u_{vz,s}\zeta_{eu,st}u'_{vz,t} \leq O_p(\frac{1}{\sqrt{Nm_{NT}}}) \quad (140)$$

Similar to the derivation of equation (124)

$$a_5 = -\frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}P_{\bar{G}}u_g u_{g,t}u'_{vz,t} \leq O_p(\frac{1}{T}) \quad (141)$$

and similar to the derivation of equation (125)

$$a_6 = \frac{1}{NT^2}\sum_{t=1}^{T}\widetilde{u}'_{vz}M_{\bar{G}}u_g\left(\widehat{u}_{g,t} - u_{g,t}\right)u'_{vz,t} \leq O_p(\frac{1}{m_{NT}}) \quad (142)$$

Therefore,

$$\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)'\widetilde{u}_{vz}=\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)'u_{vz}H+\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)'\left(\widetilde{u}_{vz}-u_{vz}H\right)\leq O_p(\frac{1}{m_{NT}}) \quad (143)$$

(3) For the term $\frac{1}{T}\widetilde{u}'_{vz}\left(\widehat{u}_{g,i}-u_{g,i}\right)$

$$\left\|\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)'P_{\bar{G}}u_{g,i}\right\|_F\leq\left(\frac{1}{T}\sum_{s=1}^{T}\left\|\widetilde{u}_{vz,s}-H'u_{vz,s}\right\|_F^2\right)^{\frac{1}{2}}\frac{1}{\sqrt{T}}\left\|P_{\bar{G}}u_{g,i}\right\|_F\leq O_p(\frac{1}{\sqrt{Tm_{NT}}}) \quad (144)$$

Similar to the derivation of equation (98)

$$\left\|\frac{1}{T}u'_{vz}P_{\bar{G}}u_{g,i}\right\|_F\leq O_p(\frac{1}{T}) \quad (145)$$

and thus

$$\frac{1}{T}\widetilde{u}'_{vz}\left(\widehat{u}_{g,i}-u_{g,i}\right)=\frac{1}{T}\left(\widetilde{u}_{vz}-u_{vz}H\right)'P_{\bar{G}}u_{g,i}+\frac{1}{T}u'_{vz}P_{\bar{G}}u_{g,i} \quad (146)$$

□

Theorem A.6. *(common component estimator)*

$$\begin{aligned}m_{NT}^{\frac{1}{2}}\left(\widetilde{c}_{it}-c_{it}\right)=&\frac{m_{NT}^{\frac{1}{2}}}{\sqrt{N}}c'_{\beta\gamma,i}\left(c'_{\beta\gamma}c_{\beta\gamma}/N\right)^{-1}\frac{1}{\sqrt{N}}\sum_{j=1}^{N}c_{\beta\gamma,j}e_{jt}\\&+\frac{m_{NT}^{\frac{1}{2}}}{\sqrt{T}}u'_{vz,t}\left(u'_{vz}u_{vz}/T\right)^{-1}\frac{1}{\sqrt{T}}u'_{vz}e_i+O_p(\frac{1}{\sqrt{m}_{NT}})\end{aligned}$$

Proof.

$$\begin{aligned}\widetilde{c}_{it}-c_{it}=&\widetilde{u}'_{vz,t}\widetilde{c}_{\beta\gamma,i}-u'_{vz,t}c_{\beta\gamma,i}=\left(\widetilde{u}_{vz,t}-H'u_{vz,t}\right)'H^{-1}c_{\beta\gamma,i}+u'_{vz,t}H\left(\widetilde{c}_{\beta\gamma,i}-H^{-1}c_{\beta\gamma,i}\right)\\&+\left(\widetilde{u}_{vz,t}-H'u_{vz,t}\right)'\left(\widetilde{c}_{\beta\gamma,i}-H^{-1}c_{\beta\gamma,i}\right)\\=&\left(\widetilde{u}_{vz,t}-H'u_{vz,t}\right)'H^{-1}c_{\beta\gamma,i}+u'_{vz,t}H\left(\widetilde{c}_{\beta\gamma,i}-H^{-1}c_{\beta\gamma,i}\right)+O_p(\frac{1}{m_{NT}})\end{aligned} \quad (147)$$

Theorem A.4 implies that

$$
\begin{aligned}
m_{NT}^{\frac{1}{2}} c'_{\beta\gamma,i} H'^{-1} \left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right) =& \frac{m_{NT}^{\frac{1}{2}}}{\sqrt{N}} c'_{\beta\gamma,i} H'^{-1} V_{NT}^{-1} \frac{1}{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} u'_{vz,s}\right) \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_{\beta\gamma,i} e_{it} + o_p(1) \\
=& \frac{m_{NT}^{\frac{1}{2}}}{\sqrt{N}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_{\beta\gamma,i} e_{it} + o_p(1) \qquad (148)
\end{aligned}
$$

and Theorem A.5 implies that

$$
\begin{aligned}
m_{NT}^{\frac{1}{2}} u'_{vz,t} H \left(\widetilde{c}_{\beta\gamma,i} - H^{-1} c_{\beta\gamma,i}\right) =& \frac{m_{NT}^{\frac{1}{2}}}{\sqrt{T}} u'_{vz,t} H H' \frac{1}{\sqrt{T}} u'_{vz} e_i + o_p(1) \\
=& \frac{m_{NT}^{\frac{1}{2}}}{\sqrt{T}} u'_{vz,t} \left(u'_{vz} u_{vz}/T\right)^{-1} \frac{1}{\sqrt{T}} u'_{vz} e_i + o_p(1) \qquad (149)
\end{aligned}
$$

where the last equality is due to

$$
HH' = \left(u'_{vz} u_{vz}/T\right)^{-1} + O_p(\frac{1}{m_{NT}}) \qquad (150)
$$

as

$$
a \qquad (151)
$$

$\left(c'_{\beta\gamma} c_{\beta\gamma}/N\right) \left(u'_{vz} \widetilde{u}_{vz}/T\right) V_{NT}^{-1}$ Therefore,

$$
\begin{aligned}
m_{NT}^{\frac{1}{2}} \left(\widetilde{c}_{it} - c_{it}\right) =& \frac{m_{NT}^{\frac{1}{2}}}{\sqrt{N}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_{\beta\gamma,i} e_{it} \\
&+ \frac{m_{NT}^{\frac{1}{2}}}{\sqrt{T}} u'_{vz,t} \left(u'_{vz} u_{vz}/T\right)^{-1} \frac{1}{\sqrt{T}} u'_{vz} e_i + O_p(\frac{1}{m_{NT}}) \qquad (152)
\end{aligned}
$$

□

Theorem A.7. *Define* $\widetilde{Q}_r = \left(\widehat{c}; \widehat{\beta}\right)$. *Then*

$$
\widetilde{Q}_r - Q_r
$$

A.5 Step (3)

$$\begin{aligned}\iota_N =& q_T\theta_g - (q - q_T)\,\theta_g \\ =& q_T\theta_{g(\bar{g},\widehat{V}_g)} + q_{T,cc}\theta_{g(\bar{g},\widehat{V}_g)} + \epsilon \end{aligned} \tag{153}$$

with

$$q_T = (c + \beta_g(\bar{g} - \mu_g), \beta_g)\,\widehat{Q}_{\bar{G}} - q_{T,cc} + \bar{e}'\bar{G}/T$$

$$\theta_{g(\bar{g},\widehat{V}_g)} = \text{diagblk}\left(1, \widehat{V}_g V_g\right)\theta_g + \left(0, \left(-\frac{\widehat{V}_g^{-1}(\bar{g} - \mu_g)}{\lambda_1}\right)'\right)' = \theta_g + \left(0, \left(-\frac{\left(\widehat{V}_g^{-1} - V_g^{-1}\right)d_g^{-1}}{\lambda_1}\lambda_f - \frac{\widehat{V}_g^{-1}(\bar{g} - \mu_g)}{\lambda_1}\right)'$$

$$\theta_g = \left(1/\lambda_1, -\frac{V_g^{-1}d_g^{-1}\lambda_f}{\lambda_1}\right)$$

$$q_{T,cc} = -\left(\beta\bar{v} + \gamma\bar{z}, \beta\frac{1}{T}\sum_{t=1}^{T} v_t\bar{g}_t' + \gamma\frac{1}{T}\sum_{t=1}^{T} z_t\bar{g}_t'\right) = -cc'\bar{G}/T$$

$$\epsilon = -\left(\bar{e}, \frac{1}{T}\sum_{t=1}^{T} e_t\bar{g}_t'\right)\theta_{g(\bar{g},\widehat{V}_g)} = -e'\bar{G}/T\theta_{g(\bar{g},\widehat{V}_g)}$$

The term $q_{T,c}$ is constructed from the common component in the error term u_g, and with preparations in steps (1) and (2) we have estimators for the common component and thus a estimator $q_{T,\widetilde{c}}$ for $q_{T,c}$

$$q_{T,\widetilde{c}} = -\left(\widetilde{c}'\iota_T/T, \widetilde{\beta}\frac{1}{T}\sum_{t=1}^{T}\widetilde{v}_t\bar{g}_t' + \widetilde{\gamma}\frac{1}{T}\sum_{t=1}^{T}\widetilde{z}_t\bar{g}_t'\right) = -\widetilde{c}'\bar{G}/T \tag{154}$$

and thus we consider

$$\iota_N = \left(q_T + q_{T,\widetilde{c}}\right)\theta_{g(\bar{g},\widehat{V}_g)} + \widetilde{\epsilon} \tag{155}$$

with $\widetilde{\epsilon} = \epsilon + \left(q_{T,c} - q_{T,\widetilde{c}}\right)\theta_{g(\bar{g},\widehat{V}_g)}$. We still have one more drawback, which is the correlation between $q_T, q_{T,\widetilde{c}}$ and $\widetilde{\epsilon}$. To solve this endogenous issue, classical approach is to use IVs. We split the whole sample into two subsamples along the time index and for each subsample we construct $q_T^{(i)}, q_{T,\widetilde{c}}^{(i)}, i = 1, 2$. Denote $\widetilde{X}_g^{(i)} = q_T^{(i)} + q_{T,\widetilde{c}}^{(i)}, i = 1, 2$. By construction $\widetilde{X}_g^{(1)}$ serve naturally as IVs

for $\widetilde{X}_g^{(2)}$, and vice versa. We consider two estimators $\widetilde{\theta}_{g,I}$ and $\widetilde{\theta}_{g,II}$.

$$\widetilde{\theta}_{g,I}^{(1)} = \mathcal{I}_{K_G}\left(\iota_N' P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)} \left(\widetilde{X}_g^{(1)\prime} P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)}\right)^{-1}\right)' \tag{156}$$

??

$$\begin{aligned}\widetilde{\theta}_{g,I}^{(1)} =& \mathcal{I}_{K_G}\left(\iota_N' P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)} \left(\widetilde{X}_g^{(1)\prime} P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)}\right)^{-1}\right)' \\ \widetilde{\theta}_{g,II} =& \mathcal{I}_{K_G}\left(\iota_N' \widetilde{Q}_r^{-1} \widetilde{B}_g \left(\widetilde{B}_g' \widetilde{Q}_r^{-1} \widetilde{B}_g\right)^{-1}\right)\end{aligned} \tag{157}$$

with $\mathcal{I}_{K_G} = \left(I_{1+K_g} \vdots 0\right)$.

Proof. 4.1

$$\widetilde{\theta}_{g,I}^{(1)} = \mathcal{I}_{K_G}\left(\iota_N' P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)} \left(\widetilde{X}_g^{(1)\prime} P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)}\right)^{-1}\right)' \tag{158}$$

Equation (155) implies

$$\widetilde{\theta}_{g,I}^{(1)} - \widetilde{\theta}_g = \left(\widetilde{X}_g^{(1)\prime} P_{\widetilde{X}_g^{(2)}} \widetilde{X}_g^{(1)}\right)^{-1} \widetilde{X}_g^{(1)} P_{\widetilde{X}_g^{(2)}} \widetilde{\epsilon}^{(1)} \tag{159}$$

$$\begin{aligned}\widetilde{X}_g^{(1)} =& (c, \beta_g)\, \widehat{Q}_g^{(1)} + \left(\widetilde{c}c^{(1)} - cc^{(1)}\right)' \bar{G}^{(1)}/|\mathcal{T}_{(1)}| + e^{(1)\prime} \bar{G}^{(1)}/|\mathcal{T}_{(1)}| \\ =& \widetilde{X}_{g,c\beta_g}^{(1)} + \widetilde{X}_{g,cc}^{(1)} + \widetilde{X}_{g,e}^{(1)}\end{aligned}$$

$$\widetilde{x}_{g,c\beta_g,i}^{(1)} = \widehat{Q}_g^{(1)} \left(c_i, \beta_{g,i}'\right)' \tag{160}$$

$$
\begin{aligned}
\widetilde{x}^{(1)}_{g,cc,i} =& \frac{1}{|\mathcal{T}_{(1)}|} \sum_{t\in\mathcal{T}_{(1)}} \left(\widetilde{cc}_{it} - cc_{it}\right) \bar{G}_t \\
=& \frac{1}{|\mathcal{T}_{(1)}|} \sum_{t\in\mathcal{T}_{(1)}} \frac{1}{\sqrt{N}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt} \bar{G}_t \\
&+ \frac{1}{|\mathcal{T}_{(1)}|} \sum_{t\in\mathcal{T}_{(1)}} \frac{1}{\sqrt{T}} u'_{vz,t} \left(u'_{vz} u_{vz}/T\right)^{-1} \frac{1}{\sqrt{T}} u'_{vz} e_i \bar{G}_t + O_p(\frac{1}{m_{NT}}) \\
=& \frac{1}{\sqrt{N|\mathcal{T}_{(1)}|}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N|\mathcal{T}_{(1)}|}} \sum_{t\in\mathcal{T}_{(1)}} \sum_{i=1}^{N} c_{\beta\gamma,i} e_{it} \bar{G}_t \\
&+ \frac{1}{\sqrt{T|\mathcal{T}_{(1)}|}} \frac{1}{\sqrt{|\mathcal{T}_{(1)}|}} \sum_{t\in\mathcal{T}_{(1)}} \bar{G}_t u'_{vz,t} \frac{1}{\sqrt{T}} \sum_{s=1}^{T} \left(u'_{vz} u_{vz}/T\right)^{-1} u_{vz,s} e_{is} + O_p(\frac{1}{m_{NT}}) \\
\leq& O_p(\frac{1}{\sqrt{m_{NT}|\mathcal{T}_{(1)}|}})
\end{aligned} \tag{161}
$$

$$
\widetilde{x}^{(1)}_{g,e,i} = \frac{1}{\sqrt{|\mathcal{T}_{(1)}|}} \frac{1}{\sqrt{|\mathcal{T}_{(1)}|}} \sum_{t\in\mathcal{T}_{(1)}} e_{it} \bar{G}'_t \tag{162}
$$

Suppose $|\mathcal{T}_{(1)}| = |\mathcal{T}_{(2)}| = \tau$. Denote

$$
Q_x = \mathrm{diag}\left(1, W_\tau\right) \tag{163}
$$

We show the following claims

$$
(1) Q_{B_g,T} \widetilde{X}^{(2)'}_g \widetilde{X}^{(2)}_g Q_{B_g,T} = \tag{164}
$$

$$
(2) \widetilde{X}^{(1)'}_g \widetilde{X}^{(2)}_g \tag{165}
$$

$$
(3) \widetilde{X}^{(2)'}_g \widetilde{\epsilon}^{(1)} \tag{166}
$$

(1)

$$\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}_{g,cc,i}^{(2)}\widetilde{x}_{g,cc,i}^{(2)\prime}\leq O_p(\frac{1}{\sqrt{\tau}}) \tag{167}$$

$$\frac{1}{N}Q_{B_g,T}\widetilde{X}_{g,c\beta_g}^{(2);}\widetilde{X}_{g,c\beta_g}^{(2)}Q_{B_g,T}\rightarrow_p Q_g\eta_{c\beta_g}Q_g \tag{168}$$

Assumption **??** implies

$$\frac{1}{N|\mathcal{T}_{(1)}|}\sum_{i=1}^{N}\sum_{t\in\mathcal{T}_{(1)}}e_{it}^2\bar{G}_t'\bar{G}_t\rightarrow_p\Sigma_{\bar{G}\bar{G}\gamma_N} \tag{169}$$

and

$$\frac{1}{\sqrt{N}}\frac{1}{\sqrt{N}|\mathcal{T}_{(1)}|}\sum_{i=1}^{N}\sum_{t\in\mathcal{T}_{(1)}}\sum_{s\neq t}e_{it}\bar{G}_t'e_{is}\bar{G}_s=O_p(\frac{1}{\sqrt{N}}) \tag{170}$$

Therefore,

$$\begin{aligned}\frac{\tau}{N}\widetilde{X}_{g,e}^{(2)\prime}\widetilde{X}_{g,e}^{(2)}&=\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}_{g,e,i}^{(2)}\widetilde{x}_{g,e,i}^{(2)\prime}=\frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}e_t\bar{G}_t'\right)\left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}e_t\bar{G}_t'\right)'\\ =&\frac{1}{N|\mathcal{T}_{(2)}|}\sum_{i=1}^{N}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t'\bar{G}_te_{it}+\frac{1}{N|\mathcal{T}_{(2)}|}\sum_{i=1}^{N}\sum_{t\in\mathcal{T}_{(2)}}\sum_{s\neq t}e_{it}\bar{G}_t'\bar{G}_se_{is}\rightarrow_p\Sigma_{\bar{G}\bar{G}\gamma_N}\end{aligned} \tag{171}$$

$$\begin{aligned}&\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}Q_{B_g,T}\widetilde{x}_{g,c\beta_g,i}^{(2)}\widetilde{x}_{g,cc,i}^{(2)\prime}\\ =&\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}}\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}\left(\widehat{Q}_g^{(1)}Q_{B_g,T}\left(c_i,\beta_{g,i}'\right)'\right)\left(c_{\beta\gamma,i}'\left(c_{\beta\gamma}'c_{\beta\gamma}/N\right)^{-1}\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}\sum_{i=1}^{N}c_{\beta\gamma,i}e_{it}\bar{G}_t\right)'\\ &+\frac{1}{\sqrt{T|\mathcal{T}_{(2)}|}}\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}\left(\widehat{Q}_g^{(2)}Q_{B_g,T}\left(c_i,\beta_{g,i}'\right)'\right)\left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}\bar{G}_tu_{vz,t}'\frac{1}{\sqrt{T}}\sum_{s=1}^{T}\left(u_{vz}'u_{vz}/T\right)^{-1}u_{vz,s}e_{is}\right)'\\ &+O_p(\frac{1}{m_{NT}})\leq O_p(\frac{1}{\sqrt{m_{NT}}})\end{aligned}$$

$$\frac{\sqrt{\tau}}{N}Q_{B_g,T}\sum_{i=1}^{N}\widetilde{x}^{(2)}_{g,c\beta_g,i}\widetilde{x}^{(2)\prime}_{g,e,i}=\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}\left(\widehat{Q}^{(1)}_{g}Q_{B_g,T}\left(c_i,\beta'_{g,i}\right)'\right)\left(\frac{1}{|\mathcal{T}_{(2)}|}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t\right)'=O_p(\frac{1}{\sqrt{N}}) \tag{172}$$

$$\begin{aligned}
&\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}^{(2)}_{g,cc,i}\widetilde{x}^{(2)\prime}_{g,e,i}\\
=&\frac{\tau}{N}\sum_{i=1}^{N}\left(\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}}c'_{\beta\gamma,i}\left(c'_{\beta\gamma}c_{\beta\gamma}/N\right)^{-1}\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}\sum_{j=1}^{N}c_{\beta\gamma,j}e_{jt}\bar{G}_t\right)\left(\frac{1}{|\mathcal{T}_{(2)}|}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t\right)'\\
&+\frac{\tau}{N}\sum_{i=1}^{N}\left(\frac{1}{\sqrt{T|\mathcal{T}_{(2)}|}}\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}\bar{G}_t u'_{vz,t}\frac{1}{\sqrt{T}}\sum_{s=1}^{T}\left(u'_{vz}u_{vz}/T\right)^{-1}u_{vz,s}e_{is}+O_p(\frac{1}{m_{NT}})\right)\\
&\times\left(\frac{1}{|\mathcal{T}_{(2)}|}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t\right)'\\
=&O_p(\frac{1}{\sqrt{\tau}})
\end{aligned} \tag{173}$$

Therefore,

$$\begin{aligned}
&\frac{1}{N}Q_{B_g,T}\widetilde{X}^{(2)\prime}_{g}\widetilde{X}^{(2)}_{g}Q_{B_g,T}=\frac{1}{N}\sum_{i=1}^{N}Q_{B_g,T}\widetilde{x}^{(2)}_{g,i}\widetilde{x}^{(2)\prime}_{g,i}Q_{B_g,T}\\
=&\frac{1}{N}\sum_{i=1}^{N}Q_{B_g,T}\left(\widetilde{x}^{(2)}_{g,c\beta_g,i}+\widetilde{x}^{(2)}_{g,cc,i}+\widetilde{x}^{(2)}_{g,e,i}\right)\left(\widetilde{x}^{(2)}_{g,c\beta_g,i}+\widetilde{x}^{(2)}_{g,cc,i}+\widetilde{x}^{(2)}_{g,e,i}\right)'Q_{B_g,T}\\
=&\frac{1}{N}Q_{B_g,T}\left(\sum_{i=1}^{N}\widetilde{x}^{(2)}_{g,c\beta_g,i}\widetilde{x}^{(2)\prime}_{g,c\beta_g,i}\right)Q_{B_g,T}+Q_{B_g,T}/\sqrt{\tau}\left(\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}^{(2)}_{g,cc,i}\widetilde{x}^{(2)\prime}_{g,cc,i}\right)Q_{B_g,T}/\sqrt{\tau}\\
&+Q_{B_g,T}/\sqrt{\tau}\left(\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}^{(2)}_{g,e,i}\widetilde{x}^{(2)\prime}_{g,e,i}\right)Q_{B_g,T}/\sqrt{\tau}+O_p(\frac{1}{\sqrt{\tau}})\\
=&Q_g\eta_{c\beta_g}Q_g+W_x\Sigma_{\bar{G}\bar{G}\gamma_N}W_x+O_p(\frac{1}{\sqrt{\tau}})
\end{aligned} \tag{174}$$

with $W_x=\lim_{\tau\to\infty}Q_{B_g,T}/\sqrt{\tau}$.

(2)

$$\frac{1}{N}Q_{B_g,T}\widetilde{X}^{(1);}_{g,c\beta_g}\widetilde{X}^{(2)}_{g,c\beta_g}Q_{B_g,T}\rightarrow_p Q_g\eta_{c\beta_g}Q_g \tag{175}$$

$$\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}^{(1)}_{g,cc,i}\widetilde{x}^{(2)\prime}_{g,cc,i}\leq O_p(\frac{1}{\sqrt{\tau}}) \tag{176}$$

$$\begin{aligned}\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}^{(1)}_{g,e,i}\widetilde{x}^{(2)\prime}_{g,e,i}&=\frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{\sqrt{|\mathcal{T}_{(1)}|}}\sum_{s\in\mathcal{T}_{(1)}}e_s\bar{G}'_s\right)\left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}e_t\bar{G}'_t\right)'\\ =&\frac{1}{N\tau}\sum_{i=1}^{N}\sum_{s\in\mathcal{T}_{(1)}}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}'_t\bar{G}_se_{is}=O_p(\frac{1}{\sqrt{N}})\end{aligned} \tag{177}$$

$$\begin{aligned}&\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}Q_{B_g,T}\widetilde{x}^{(1)}_{g,c\beta_g,i}\widetilde{x}^{(2)\prime}_{g,cc,i}\\ =&\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}}\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}\left(\widehat{Q}^{(1)}_gQ_{B_g,T}\left(c_i,\beta'_{g,i}\right)'\right)\left(c'_{\beta\gamma,i}\left(c'_{\beta\gamma}c_{\beta\gamma}/N\right)^{-1}\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}\sum_{i=1}^{N}c_{\beta\gamma,i}e_{it}\bar{G}_t\right)'\\ &+\frac{1}{\sqrt{T|\mathcal{T}_{(2)}|}}\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}\left(\widehat{Q}^{(1)}_gQ_{B_g,T}\left(c_i,\beta'_{g,i}\right)'\right)\left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}}\sum_{t\in\mathcal{T}_{(2)}}\bar{G}_tu'_{vz,t}\frac{1}{\sqrt{T}}\sum_{s=1}^{T}\left(u'_{vz}u_{vz}/T\right)^{-1}u_{vz,s}e_{is}\right)'\\ &+O_p(\frac{1}{\sqrt{m_{NT}}})=O_p(\frac{1}{\sqrt{m_{NT}}})\end{aligned}$$

$$\frac{\sqrt{\tau}}{N}Q_{B_g,T}\sum_{i=1}^{N}\widetilde{x}^{(1)}_{g,c\beta_g,i}\widetilde{x}^{(2)\prime}_{g,e,i}=\frac{\sqrt{\tau}}{N}\sum_{i=1}^{N}\left(\widehat{Q}^{(1)}_gQ_{B_g,T}\left(c_i,\beta'_{g,i}\right)'\right)\left(\frac{1}{|\mathcal{T}_{(2)}|}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t\right)'=O_p(\frac{1}{\sqrt{N}}) \tag{178}$$

$$
\begin{aligned}
&\frac{\tau}{N}\sum_{i=1}^{N}\widetilde{x}_{g,cc,i}^{(1)}\widetilde{x}_{g,e,i}^{(2)\prime}\\
=&\frac{\tau}{N}\sum_{i=1}^{N}\left(\frac{1}{\sqrt{N|\mathcal{T}_{(1)}|}}c_{\beta\gamma,i}'\left(c_{\beta\gamma}'c_{\beta\gamma}/N\right)^{-1}\frac{1}{\sqrt{N|\mathcal{T}_{(1)}|}}\sum_{t\in\mathcal{T}_{(1)}}\sum_{j=1}^{N}c_{\beta\gamma,j}e_{jt}\bar{G}_t\right)\left(\frac{1}{|\mathcal{T}_{(2)}|}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t\right)'\\
&+\frac{\tau}{N}\sum_{i=1}^{N}\left(\frac{1}{\sqrt{T|\mathcal{T}_{(1)}|}}\frac{1}{\sqrt{|\mathcal{T}_{(1)}|}}\sum_{t\in\mathcal{T}_{(1)}}\bar{G}_t u_{vz,t}'\frac{1}{\sqrt{T}}\sum_{s=1}^{T}\left(u_{vz}'u_{vz}/T\right)^{-1}u_{vz,s}e_{is}+O_p(\frac{1}{m_{NT}})\right)\\
&\times\left(\frac{1}{|\mathcal{T}_{(2)}|}\sum_{t\in\mathcal{T}_{(2)}}e_{it}\bar{G}_t\right)'\\
=&O_p(\frac{1}{\sqrt{N}})+O_p(\frac{1}{\sqrt{\tau}})
\end{aligned}
\tag{179}
$$

Therefore,

$$
\begin{aligned}
&\frac{1}{N}Q_{B_g,T}\widetilde{X}_g^{(1)\prime}\widetilde{X}_g^{(2)}Q_{B_g,T}=\frac{1}{N}\sum_{i=1}^{N}Q_{B_g,T}\widetilde{x}_{g,i}^{(1)}\widetilde{x}_{g,i}^{(2)\prime}Q_{B_g,T}\\
=&\frac{1}{N}\sum_{i=1}^{N}Q_{B_g,T}\left(\widetilde{x}_{g,c\beta_g,i}^{(1)}+\widetilde{x}_{g,cc,i}^{(1)}+\widetilde{x}_{g,e,i}^{(1)}\right)\left(\widetilde{x}_{g,c\beta_g,i}^{(2)}+\widetilde{x}_{g,cc,i}^{(2)}+\widetilde{x}_{g,e,i}^{(2)}\right)'Q_{B_g,T}\\
=&\frac{1}{N}Q_{B_g,T}\left(\sum_{i=1}^{N}\widetilde{x}_{g,c\beta_g,i}^{(1)}\widetilde{x}_{g,c\beta_g,i}^{(2)\prime}\right)Q_{B_g,T}+O_p(\frac{1}{\sqrt{\tau}})=Q_g\eta_{c\beta_g}Q_g+O_p(\frac{1}{\sqrt{\tau}})
\end{aligned}
\tag{180}
$$

(3)

$$
\widetilde{\epsilon}_i^{(1)}=\left(\widetilde{x}_{g,e,i}^{(1)}+\widetilde{x}_{g,cc,i}^{(1)}\right)\theta_{g(\bar{g},\widehat{V}_g)}
\tag{181}
$$

Equations (177)-(179) imply that

$$
\begin{aligned}
&\frac{\sqrt{\tau}}{\sqrt{N}}Q_{B_g,T}\sum_{i=1}^{N}\widetilde{x}_{g,i}^{(2)}\left(\widetilde{x}_{g,e,i}^{(1)}+\widetilde{x}_{g,cc,i}^{(1)}\right)'\\
=&\frac{\sqrt{\tau}}{\sqrt{N}}Q_{B_g,T}\sum_{i=1}^{N}\left(\widetilde{x}_{g,c\beta_g,i}^{(2)}+\widetilde{x}_{g,cc,i}^{(2)}+\widetilde{x}_{g,e,i}^{(2)}\right)\left(\widetilde{x}_{g,e,i}^{(1)}+\widetilde{x}_{g,cc,i}^{(1)}\right)'=O_p(1)
\end{aligned}
\tag{182}
$$

which provided that $\|\theta_g\|_F\leq L$ (here we need to discuss the assumption in Kleibergen and Zhan

(2015)) then gives

$$\sqrt{\frac{\tau}{N}} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{\epsilon}^{(1)} = O_p(1) \tag{183}$$

Finally

$$\begin{aligned}
&\sqrt{N\tau} Q_{B_g,T}^{-1} \left(\widetilde{\theta}_{g,I}^{(1)} - \widetilde{\theta}_g \right) \\
=& \left(\frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(1)\prime} \widetilde{X}_g^{(2)} Q_{B_g,T} \left(\frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{X}_g^{(2)} Q_{B_g,T} \right)^{-1} \frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{X}_g^{(1)} Q_{B_g,T} \right)^{-1} \\
&\times \frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(1)} \widetilde{X}_g^{(2)} Q_{B_g,T} \left(\frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{X}_g^{(2)} Q_{B_g,T} \right)^{-1} \sqrt{\frac{\tau}{N}} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{\epsilon}^{(1)} = O_p(1)
\end{aligned} \tag{184}$$

Next we need to show the asymptotic distribution of the $\widetilde{\theta}_g$. From the above discussions, we know

$$\begin{aligned}
&\left(\frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(1)\prime} \widetilde{X}_g^{(2)} Q_{B_g,T} \left(\frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{X}_g^{(2)} Q_{B_g,T} \right)^{-1} \frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{X}_g^{(1)} Q_{B_g,T} \right)^{-1} \\
&\times \frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(1)} \widetilde{X}_g^{(2)} Q_{B_g,T} \left(\frac{1}{N} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{X}_g^{(2)} Q_{B_g,T} \right)^{-1} \to_p \Theta_{(1)}
\end{aligned} \tag{185}$$

with $\Theta_{(1)}$ a deterministic positive definite matrix. We then need to derive the asymptotic distribution of the term $\sqrt{\frac{\tau}{N}} Q_{B_g,T} \widetilde{X}_g^{(2)\prime} \widetilde{\epsilon}^{(1)}$.

$$\widetilde{x}_{g,c\beta_g,i}^{(2)} = \widehat{Q}_g^{(2)} \left(c_i, \beta_{g,i}' \right)' \tag{186}$$

Equation (147) implies that

$$
\begin{aligned}
\widetilde{c}_{it} - c_{it} =& \left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right)' H^{-1} c_{\beta\gamma,i} + u'_{vz,t} H \left(\widetilde{c}_{\beta\gamma,i} - H^{-1} c_{\beta\gamma,i}\right) \\
&+ \left(\frac{1}{\sqrt{N}} V_{NT}^{-1} \frac{1}{T} \sum_{s=1}^{T} \left(\widetilde{u}_{vz,s} u'_{vz,s}\right) \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_{\beta\gamma,i} e_{it}\right)' \left(\frac{1}{\sqrt{T}} H' \frac{1}{\sqrt{T}} u'_{vz} e_i\right) + o_p(\frac{1}{m_{NT}}) \\
=& \left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right)' H^{-1} c_{\beta\gamma,i} + u'_{vz,t} H \left(\widetilde{c}_{\beta\gamma,i} - H^{-1} c_{\beta\gamma,i}\right) \\
&+ \left(\frac{1}{T} u'_{vz} e_i\right)' \left(\frac{1}{\sqrt{N}} H H' \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt}\right) + o_p(\frac{1}{m_{NT}}) \\
=& \left(\widetilde{u}_{vz,t} - H' u_{vz,t}\right)' H^{-1} c_{\beta\gamma,i} + u'_{vz,t} H \left(\widetilde{c}_{\beta\gamma,i} - H^{-1} c_{\beta\gamma,i}\right) \\
&+ \left(\frac{1}{T} u'_{vz} e_i\right)' \left(\frac{1}{\sqrt{N}} \left(u'_{vz} u_{vz}/T\right)^{-1} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt}\right) + o_p(\frac{1}{m_{NT}}) \\
=& \frac{1}{\sqrt{N}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt} + \frac{1}{\sqrt{T}} u'_{vz,t} \left(u'_{vz} u_{vz}/T\right)^{-1} \frac{1}{\sqrt{T}} u'_{vz} e_i \\
&+ \left(\frac{1}{T} u'_{vz} e_i\right)' \left(\frac{1}{\sqrt{N}} \left(u'_{vz} u_{vz}/T\right)^{-1} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt}\right) + o_p(\frac{1}{m_{NT}}) \quad (187)
\end{aligned}
$$

and thus

$$
\begin{aligned}
\widetilde{x}^{(2)}_{g,cc,i} =& \frac{1}{|\mathcal{T}_{(2)}|} \sum_{t \in \mathcal{T}_{(2)}} \frac{1}{\sqrt{N}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt} \bar{G}_t \\
&+ \frac{1}{|\mathcal{T}_{(2)}|} \sum_{t \in \mathcal{T}_{(2)}} \frac{1}{\sqrt{T}} u'_{vz,t} \left(u'_{vz} u_{vz}/T\right)^{-1} \frac{1}{\sqrt{T}} u'_{vz} e_i \bar{G}_t \\
&+ \frac{1}{|\mathcal{T}_{(2)}|} \sum_{t \in \mathcal{T}_{(2)}} \left(\frac{1}{T} u'_{vz} e_i\right)' \frac{1}{\sqrt{N}} \left(u'_{vz} u_{vz}/T\right)^{-1} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} e_{jt} \bar{G}_t + o_p(\frac{1}{m_{NT}}) \\
=& \frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}} \left(\frac{1}{\sqrt{N|\mathcal{T}_{(2)}|}} \sum_{j=1}^{N} \sum_{t \in \mathcal{T}_{(2)}} c'_{\beta\gamma,i} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} c_{\beta\gamma,j} e_{jt} \bar{G}_t\right) \\
&+ \frac{1}{\sqrt{T|\mathcal{T}_{(2)}|}} \left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}} \sum_{t \in \mathcal{T}_{(2)}} \bar{G}_t u'_{vz,t}\right) \left(u'_{vz} u_{vz}/T\right)^{-1} \left(\frac{1}{\sqrt{T}} u'_{vz} e_i\right) \\
&+ \frac{1}{\sqrt{|\mathcal{T}_{(2)}|}} \left(\frac{1}{T} u'_{vz} e_i\right)' \frac{1}{\sqrt{N}} \left(u'_{vz} u_{vz}/T\right)^{-1} \left(c'_{\beta\gamma} c_{\beta\gamma}/N\right)^{-1} \frac{1}{\sqrt{N}} \sum_{j=1}^{N} c_{\beta\gamma,j} \left(\frac{1}{\sqrt{|\mathcal{T}_{(2)}|}} \sum_{t \in \mathcal{T}_{(2)}} e_{jt} \bar{G}_t\right) \\
&+ o_p(\frac{1}{m_{NT}}) = O_p(\frac{1}{\sqrt{NT}}) + O_p(\frac{1}{T}) + O_p(\frac{1}{\sqrt{N}T}) + o_p(\frac{1}{m_{NT}}) \quad (188)
\end{aligned}
$$

$$\widetilde{x}^{(2)}_{g,e,i} = \frac{1}{\sqrt{|\mathcal{T}_{(2)}|}} \frac{1}{\sqrt{|\mathcal{T}_{(2)}|}} \sum_{t \in \mathcal{T}_{(2)}} e_{it} \bar{G}'_t \tag{189}$$

$$\frac{T}{N} \sum_{i=1}^{N} Q_{B_g,T} \widetilde{x}^{(2)}_{g,i} \widehat{\epsilon}^{(1)\prime}_i \widetilde{\epsilon}^{(1)}_i \widetilde{x}^{(2)\prime}_{g,i} Q_{B_g,T} \to_p \widetilde{\Sigma} \tag{190}$$

Our estimate $\widetilde{\theta}^{(1)}_G$ then resulted from the following regression with $\widetilde{q}^{(2)}_{G,T}$ instrumented for $\widetilde{q}^{(1)}_{G,T}$

$$\iota_N = \widetilde{q}^{(1)}_{G,T} \widetilde{\theta}^{(1)}_{G,T} - \widetilde{\epsilon}^{(1)}_{q_G} \widetilde{\theta}^{(1)}_{G,T},$$

with

$$\widetilde{\epsilon}^{(1)}_{q_G} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} (e_t + cc_t - \widetilde{c}c_t)\, G'_t, \widehat{V}_{g^{(1)}} = \frac{1}{\lfloor \frac{T}{2} \rfloor} \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} \bar{g}_t \bar{g}'_t,$$

$$\widetilde{\theta}^{(1)}_{G,T} = \theta_G + O_p(1/\sqrt{T}) = \theta_G + \left(0, \left(-\left(\widehat{V}^{-1}_{g^{(1)}} - V^{-1}_g\right) d^{-1}_g \lambda_f / \lambda_1 - \widehat{V}^{-1}_{g^{(1)}} (\bar{g} - \mu_g)/\lambda_1\right)'\right)'.$$

□